

Supplemental Materials for Conditional Relationships in Dynamic Models

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1 Extension to the ECM framework

The ADL model from Equation 3 in the main text translates into an equivalent ECM:

$$\begin{aligned}\Delta y_t = & \gamma_0 + \gamma_1 y_{t-1} + \theta_0 \Delta x_t + \theta_1 x_{t-1} + \theta_2 \Delta z_t + \theta_3 z_{t-1} + \\ & \theta_4 \Delta x_t z_{t-1} + \theta_5 x_{t-1} \Delta z_t + \theta_6 \Delta x_t \Delta z_t + \theta_7 x_{t-1} z_{t-1} + \epsilon_t.\end{aligned}\tag{SM.1}$$

This equivalence has recently been brought under scrutiny by (Grant and Lebo, 2016), who argue that (compared to the ADL) the ECM directly estimates different quantities, finds significant results more routinely, and thus makes mistaken inferences more likely. Yet this misdiagnoses the problem. The ECM leads to inferential errors not because of the parameterization of the model *per se*, but rather because scholars pay too little attention to appropriately deriving and interpreting quantities of interest. See (Keele et al., 2016) for simulation evidence on their equivalence, and (Enns et al., 2016) for further discussion.

We provide the variable translations to move between the ADL and ECM frameworks when we discuss quantities of interest in Section SM 4.3. We still need to establish the appropriateness of the ECM as a modeling framework for conditional interactions. Accordingly, we analyze the case in which the variables are cointegrated.

Proposition 2. If a cointegrating vector exists for a dynamic system composed of a response y and two covariates x and z , then a cointegrating vector also exists for a system including the multiplicative interaction xz .

This Proposition states that if the variables in the model are cointegrated, then adding an interaction term makes no further demands of the data. OLS estimates will still be consistent. Moreover, this condition is only sufficient, not necessary: it is possible that the dynamic system *becomes* cointegrated when an interaction is included. Intuitively, adding an interaction may actually induce equation balance in an otherwise-unbalanced model. Appropriately modeling conditional relationships may uncover cointegrating vectors and

allow scholars to reliably estimate dynamic effects for which OLS would not otherwise be consistent.

As with all ADLs (Philips, 2022), scholars should still explicitly test models with a conditional relationship for cointegration. A number of diagnostics already exist, such as the Engle and Granger (1987), Johansen (1988), and Pesaran et al. (2001) “bounds” tests (for an implementation, see Philips (2018)). But adding a conditional relationship should feel like a natural extension of tests that are familiar to the researcher. Generally, though, we leave the issue of the complexity of cointegration in the dynamic interaction setting, especially using tests for cointegration, to future work.

2 Proofs

2.1 Proposition 1

Proof. Recall that for the model of interest there are only two covariates, x and z , and a response y , each of which is a covariance-stationary AR(1) process. These time series can then be represented individually as

$$y_t = \zeta_y + \rho_y y_{t-1} + \eta_t, \quad \text{where } |\rho_y| < 1, \quad (\text{SM.2})$$

$$x_t = \zeta_x + \rho_x x_{t-1} + \eta_t, \quad \text{where } |\rho_x| < 1, \quad (\text{SM.3})$$

$$z_t = \zeta_z + \rho_z z_{t-1} + \eta_t, \quad \text{where } |\rho_z| < 1, \quad (\text{SM.4})$$

with t indexing time. Assume that $\eta \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$. Following Davidson and MacKinnon (2003) and Box-Steffensmeier et al. (2014), the unconditional mean and variance, and unconditional autocovariance for any lag length ℓ , can be written

$$\mathbb{E}[y_t] = \frac{\zeta_y}{1 - \rho_y}, \quad \mathbb{V}[y_t] = \frac{\sigma^2}{1 - \rho_y^2}, \quad \mathbb{C}[y_t, y_{t+\ell}] = \rho_y^\ell \left(\frac{\sigma^2}{1 - \rho_y^2} \right), \quad (\text{SM.5})$$

$$\mathbb{E}[x_t] = \frac{\zeta_x}{1 - \rho_x}, \quad \mathbb{V}[x_t] = \frac{\sigma^2}{1 - \rho_x^2}, \quad \mathbb{C}[x_t, x_{t+\ell}] = \rho_x^\ell \left(\frac{\sigma^2}{1 - \rho_x^2} \right), \quad (\text{SM.6})$$

$$\mathbb{E}[z_t] = \frac{\zeta_z}{1 - \rho_z}, \quad \mathbb{V}[z_t] = \frac{\sigma^2}{1 - \rho_z^2}, \quad \mathbb{C}[z_t, z_{t+\ell}] = \rho_z^\ell \left(\frac{\sigma^2}{1 - \rho_z^2} \right). \quad (\text{SM.7})$$

Define $\mathcal{F}(\mu, \sigma^2)$ as generic distribution with mean μ , variance σ^2 , and autocovariance $\gamma(\ell)$ for lag ℓ . Thus, we can define the variables as stochastic processes using the Equations

SM.5-SM.7 as

$$y \sim \mathcal{F}(\mu_y, \sigma_y^2), \quad (\text{SM.8})$$

$$x \sim \mathcal{F}(\mu_x, \sigma_x^2), \quad (\text{SM.9})$$

$$z \sim \mathcal{F}(\mu_z, \sigma_z^2). \quad (\text{SM.10})$$

We now calculate the unconditional mean, variance, and autocovariance of $x_t z_t$ in turn.

Mean. The expectation is given by

$$\begin{aligned} \mathbb{E}[x_t z_t] &= \mathbb{C}[x_t, z_t] + \mathbb{E}[x_t] \mathbb{E}[z_t] \\ &= \mathbb{C}[x_t, z_t] + \mu_x \mu_z. \end{aligned} \quad (\text{SM.11})$$

Variance. The variance is given by

$$\begin{aligned} \mathbb{V}[x_t z_t] &= (\mathbb{E}[x_t])^2 \mathbb{V}[z_t] + (\mathbb{E}[z_t])^2 \mathbb{V}[x_t] + \mathbb{E}[(x_t - \mathbb{E}[x_t])^2 (z_t - \mathbb{E}[z_t])^2] \\ &\quad + 2\mathbb{E}[x_t] \mathbb{E}[(x_t - \mathbb{E}[x_t])(z_t - \mathbb{E}[z_t])^2] \\ &\quad + 2\mathbb{E}[z_t] \mathbb{E}[(x_t - \mathbb{E}[x_t])^2 (z_t - \mathbb{E}[z_t])] \\ &\quad + 2\mathbb{E}[x_t] \mathbb{E}[z_t] \mathbb{C}[x_t, z_t] - (\mathbb{C}[x_t, z_t])^2, \end{aligned}$$

allowing for

$$\begin{aligned} (\mathbb{E}[x_t])^2 \mathbb{V}[z_t] &= \mu_x^2 \sigma_z^2 \\ (\mathbb{E}[z_t])^2 \mathbb{V}[x_t] &= \mu_z^2 \sigma_x^2 \\ \mathbb{E}[(x_t - \mathbb{E}[x_t])^2 (z_t - \mathbb{E}[z_t])^2] &= \mathbb{C}(x_t^2, z_t^2) \end{aligned}$$

this quantity reduces to

$$\begin{aligned}\mathbb{V}[x_t z_t] &= \mathbb{C}[x_t^2, z_t^2] + 2\mu_x \mathbb{C}[x_t, z_t^2] + 2\mu_z \mathbb{C}[x_t^2, z_t] - (\mathbb{C}[x_t, z_t])^2 \\ &\quad + (\sigma_x^2 + 2\mu_x^2 - 2\mu_x)(\sigma_z^2 + 2\mu_z^2 - 2\mu_z).\end{aligned}\tag{SM.12}$$

Auto-covariance. Bohrnstedt and Goldberger (1969) build on results from Goodman (1960), giving the general form of the covariance of two products of random variables:

$$\begin{aligned}\mathbb{C}[x_t z_t, x_{t+\ell} z_{t+\ell}] &= \\ &\quad \mathbb{E}[x_t] \mathbb{E}[x_{t+\ell}] \mathbb{C}[z_t, z_{t+\ell}] + \mathbb{E}[x_t] \mathbb{E}[z_{t+\ell}] \mathbb{C}[x_{t+\ell}, z_t] \\ &\quad + \mathbb{E}[z_t] \mathbb{E}[x_{t+\ell}] \mathbb{C}[x_t, z_{t+\ell}] + \mathbb{E}[z_t] \mathbb{E}[z_{t+\ell}] \mathbb{C}[x_t, x_{t+\ell}] \\ &\quad + \mathbb{E}[(x_t - \mathbb{E}[x_t])(z_t - \mathbb{E}[z_t])(x_{t+\ell} - \mathbb{E}[x_{t+\ell}](z_{t+\ell} - \mathbb{E}[z_{t+\ell}]))] \\ &\quad + \mathbb{E}[x_t] \mathbb{E}[(z_t - \mathbb{E}[z_t])(x_{t+\ell} - \mathbb{E}[x_{t+\ell}](z_{t+\ell} - \mathbb{E}[z_{t+\ell}]))] \\ &\quad + \mathbb{E}[z_t] \mathbb{E}[(x_t - \mathbb{E}[x_t])(x_{t+\ell} - \mathbb{E}[x_{t+\ell}](z_{t+\ell} - \mathbb{E}[z_{t+\ell}]))] \\ &\quad + \mathbb{E}[x_{t+\ell}] \mathbb{E}[(x_t - \mathbb{E}[x_t])(z_t - \mathbb{E}[z_t])(z_{t+\ell} - \mathbb{E}[z_{t+\ell}]))] \\ &\quad + \mathbb{E}[z_{t+\ell}] \mathbb{E}[(x_t - \mathbb{E}[x_t])(z_t - \mathbb{E}[z_t])(x_{t+\ell} - \mathbb{E}[x_{t+\ell}]))] \\ &\quad - \mathbb{C}[x_t, z_t] \mathbb{C}[x_{t+\ell}, z_{t+\ell}].\end{aligned}\tag{SM.13}$$

Making use of the same general identities, this expression simplifies to

$$\begin{aligned}\mathbb{C}[x_t z_t, x_{t+\ell} z_{t+\ell}] &= \\ &\quad 4\mu_x^2 \mu_z^2 + \mu_x^2 \rho_z^\ell \sigma_z^2 + \mu_z^2 \rho_x^\ell \sigma_x^2 + \rho_x^\ell \rho_z^\ell \sigma_x^2 \sigma_z^2 \\ &\quad + \mu_x \left(\mathbb{C}[x_t, z_t z_{t+\ell}] - \mathbb{C}[x_{t+\ell}, z_t z_{t+\ell}] \right) + \mu_z \left(\mathbb{C}[x_t x_{t+\ell}, z_t] - \mathbb{C}[x_t x_{t+\ell}, z_{t+\ell}] \right) \\ &\quad - \mu_x \mu_z \left(\mathbb{C}[x_t, z_t] + \mathbb{C}[x_{t+\ell}, z_{t+\ell}] \right) + \mathbb{C}[x_t x_{t+\ell}, z_t z_{t+\ell}] + \mathbb{C}[x_t, z_t] \mathbb{C}[x_{t+\ell}, z_{t+\ell}].\end{aligned}\tag{SM.14}$$

Checking for stationarity. Recall that the definition of weak (covariance) stationarity is that, for a stochastic series, the unconditional mean and variance, and unconditional

autocovariance for some lag ℓ , must exist and be independent of t (for $t, \ell \in \mathbb{Z}$).

The stochastic process xz satisfies these conditions only where the covariance terms in Equations SM.11, SM.12, and SM.14 are not functions of time. Thus, $x_t z_t$ will be stationary if and only if $\mathbb{C}[\hat{x}, \hat{z}] = \mathbb{C}[\tilde{x}, \tilde{z}] \forall \hat{x}$ and $\tilde{x} \in \{x_t, x_{t+\ell}, x_t x_{t+\ell}\}, \hat{z}$ and $\tilde{z} \in \{z_t, z_{t+\ell}, z_t z_{t+\ell}\}$, where $t, \ell \in \mathbb{Z}_{\geq 0}$, and $\hat{x}$ ($\hat{z}$) and $\tilde{x}$ ($\tilde{z}$) differ only in t . ■

2.2 Corollary 1

Proof. Assume x and z are stochastically independent. We calculate the the mean, variance, and auto-covariance of $x_t z_t$ in turn.

Mean. The mean is given by

$$\begin{aligned} \mathbb{E}[x_t z_t] &= \mathbb{E}[x_t] \mathbb{E}[z_t] \\ &= \mu_x \mu_z. \end{aligned} \tag{SM.15}$$

Variance. The variance is given by

$$\begin{aligned} \mathbb{V}[x_t z_t] &= (\mathbb{E}[x_t])^2 \mathbb{V}(z_t) + (\mathbb{E}[z_t])^2 \mathbb{V}(x_t) + \mathbb{V}(z_t) \mathbb{V}(x_t) \\ &= \mu_x^2 \sigma_z^2 + \mu_z^2 \sigma_x^2 + \sigma_x^2 \sigma_z^2. \end{aligned} \tag{SM.16}$$

Autocovariance. For independent x and z , all of the third moments in Equation SM.13 reduce to zero, and the fourth moment reduces to the product of each variable's autocovari-

ance, such that

$$\begin{aligned}
\mathbb{C}[x_t z_t, x_{t+\ell} z_{t+\ell}] &= \mathbb{E}[x_t] \mathbb{E}[x_{t+\ell}] \mathbb{C}[z_t, z_{t+\ell}] + \mathbb{E}[x_t] \mathbb{E}[z_{t+\ell}] \mathbb{C}[x_{t+\ell}, z_t] \\
&\quad + \mathbb{E}[z_t] \mathbb{E}[x_{t+\ell}] \mathbb{C}[x_t, z_{t+\ell}] + \mathbb{E}[z_t] \mathbb{E}[z_{t+\ell}] \mathbb{C}[x_t, x_{t+\ell}] \\
&\quad + \rho_x^\ell \rho_z^\ell \sigma_x^2 \sigma_z^2 - \mathbb{C}[x_t, z_t] \mathbb{C}[x_{t+\ell}, z_{t+\ell}] \\
&= \mu_x^2 \rho_z^\ell \sigma_z^2 + \mu_z^2 \rho_x^\ell \sigma_x^2 + \rho_x^\ell \rho_z^\ell \sigma_x^2 \sigma_z^2.
\end{aligned} \tag{SM.17}$$

Checking for stationarity. The unconditional mean is constant, all of the terms in the unconditional variance are positive constants, and all of the terms in the autocovariance are constant. Thus the stochastic process xz is stationary and can be described:

$$\begin{aligned}
xz &\sim \mathcal{F}(\mu_x \mu_z, \mu_x^2 \sigma_z^2 + \mu_z^2 \sigma_x^2 + \sigma_x^2 \sigma_z^2), \\
\gamma_{xz}(\ell) &= \mu_x^2 \rho_z^\ell \sigma_z^2 + \mu_z^2 \rho_x^\ell \sigma_x^2 + \rho_x^\ell \rho_z^\ell \sigma_x^2 \sigma_z^2.
\end{aligned} \quad \blacksquare$$

2.3 Proposition 2

Proof. Recall that a cointegrating vector is a vector of coefficients such that the linear combination of more than one stochastic series produces a stationary stochastic process, where at least one element of the vector is non-zero. Assume y , x , and z are AR(1) stochastic processes. Define the trivariate system with no multiplicative interactions among covariates x and z as

$$y_t - \alpha_0 - \alpha_1 y_{t-1} - \beta_0 x_t - \beta_1 x_{t-1} - \beta_2 z_t - \beta_3 z_{t-1} = \epsilon_t, \tag{SM.18}$$

where $\epsilon \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$. Assume that there is at least one cointegrating vector for this system; without loss of generality, define this cointegrating vector in the normalized form $\beta = (1, -\alpha_1, -\beta_0, -\beta_1, -\beta_2, -\beta_3)$.

Define the system that includes a multiplicative interaction between x_t and z_t as

$$y_t - \alpha_0 - \alpha_1 y_{t-1} - \beta_0 x_t - \beta_1 x_{t-1} - \beta_2 z_t - \beta_3 z_{t-1} - \beta_4 x_t z_t = \epsilon_t. \quad (\text{SM.19})$$

Consider the candidate cointegrating vector

$$\beta' = (1, -\alpha_1, -\beta_0, -\beta_1, -\beta_2, -\beta_3, -\beta_4),$$

where $\beta_4 = 0$. It is evident that this produces a stationary stochastic process ϵ , by the assumption that β does: β_4 makes the multiplicative interaction zero, and by assumption, the remaining coefficients produce a linear combination of y , x , and z such that ϵ is stationary. Further, the assumption that β is a cointegrating vector implies that at least one term in β —and therefore β' —is non-zero. Thus β' is a cointegrating vector, and the system with a multiplicative interaction among covariates is cointegrated. ■

3 Extension to an AR(p) process

Proof. The response y and covariates x and z can be defined as stationary AR(p) processes:

$$y_t = \zeta_y + \rho_{y,1}y_{t-1} + \rho_{y,2}y_{t-2} + \cdots + \rho_{y,p}y_{t-p} + \eta_t \quad (\text{SM.20})$$

$$x_t = \zeta_x + \rho_{x,1}x_{t-1} + \rho_{x,2}x_{t-2} + \cdots + \rho_{x,p}x_{t-p} + \eta_t \quad (\text{SM.21})$$

$$z_t = \zeta_z + \rho_{z,1}z_{t-1} + \rho_{z,2}z_{t-2} + \cdots + \rho_{z,p}z_{t-p} + \eta_t. \quad (\text{SM.22})$$

Allow L to denote the lag operator—when L multiplies any parameter with a time subscript, this subscript is lagged one period—and define

$$\phi(L) = 1 - \rho_1 L - \rho_2 L^2 - \cdots - \rho_p L^p. \quad (\text{SM.23})$$

Stationarity requires that the roots of the polynomial equation $\phi(L) = 0$ lie outside the unit circle for y , x , and z . This allows us to rewrite the unconditional mean, unconditional variance, and autocovariance conditional on ℓ as

$$\mathbb{E}[y_t] = \frac{\zeta_y}{1 - \sum_{i=1}^p \rho_{y,i}}, \quad \mathbb{V}[y_t] = \frac{\sigma^2}{1 - \sum_{i=1}^p \rho_{y,i}^2}, \quad \mathbb{C}[y_t, y_{t+\ell}] = \sum_{r=1}^p \rho_{y,r} \gamma_y(\ell - r) \quad (\text{SM.24})$$

$$\mathbb{E}[x_t] = \frac{\zeta_x}{1 - \sum_{i=1}^p \rho_{x,i}}, \quad \mathbb{V}[x_t] = \frac{\sigma^2}{1 - \sum_{i=1}^p \rho_{x,i}^2}, \quad \mathbb{C}[x_t, x_{t+\ell}] = \sum_{r=1}^p \rho_{x,r} \gamma_x(\ell - r) \quad (\text{SM.25})$$

$$\mathbb{E}[z_t] = \frac{\zeta_z}{1 - \sum_{i=1}^p \rho_{z,i}}, \quad \mathbb{V}[z_t] = \frac{\sigma^2}{1 - \sum_{i=1}^p \rho_{z,i}^2}, \quad \mathbb{C}[z_t, z_{t+\ell}] = \sum_{r=1}^p \rho_{z,r} \gamma_z(\ell - r) \quad (\text{SM.26})$$

As above, define the mean, variance, and autocorrelation of y as μ_y , σ_y^2 , and $\gamma_y(\ell)$, respectively, and similarly for x and z . The same result as in the AR(1) case holds. $\blacksquare$

4 Quantities of interest

4.1 Generally defined

We suggest three potential quantities of interest for scholars of dynamic models to investigate: period-specific effects, cumulative effects, and threshold effects.

4.1.1 Period-specific effects

The first of these quantities are period-specific effects, defined as the change in y at any time as a result of a shock to x at time t . In the main text, we focus on the case in which $j = 0$, but differentiating with respect to x_t yields the general expression for period-specific effects:

$$\frac{\partial y_{t+j}}{\partial x_t} = \begin{cases} 0 & \text{for } j \in \mathbb{Z}_{<0}, \\ \beta_0 + \beta_4 z_t + \beta_6 z_{t-1} & \text{for } j = 0, \\ \alpha_1^j (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) + \alpha_1^{j-1} (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t) & \text{for } j \in \mathbb{Z}_{>0}. \end{cases} \quad (\text{SM.27})$$

This is an intuitive result: the effect of the shock is transmitted through the coefficient on each x_t term, decaying at a rate determined by the error correction rate, α_1 . Since x is itself autoregressive, the shock also decays through its own lagged values, weighted by exactly α_1 less. All of these parameters are directly estimated, so any period-specific effect is straightforward to calculate from the model output.

Scholars typically restrict attention to the case where $j = 0$, which De Boef and Keele (2008) refer to as the “short-run effect.” Yet this quantity—better understood as the *instantaneous* effect of x on y —is only one among many period-specific effects that may be useful for studying a dynamic system. For example, Casillas et al. (2011) examine the effect of public opinion on Supreme Court decisions. Computing period-specific effects from their estimates reveals that the instantaneous effect at time t is 1.59, the effect in period $t + 1$

is -0.42, in period $t + 2$ it is -0.17, and so on. Rather than monotonically increasing over time, the effect of public opinion on judicial decisions is immediately large but diminishes significantly. Studying period-specific effects can uncover nuanced relationships, improving our understanding of dynamic systems.

4.1.2 Cumulative effects

The second of these quantities are cumulative effects: over the course of some period, what is the total change in y as a result of a shock to x ? We can imagine this as a sum of period-specific effects: for any time window $[t + h, t + k]$, the total change in y as the result of a shock to x is

$$\sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} \equiv \frac{\partial y_{t+h}}{\partial x_t} + \frac{\partial y_{t+h+1}}{\partial x_t} + \dots + \frac{\partial y_{t+k}}{\partial x_t}, \quad (\text{SM.28})$$

where h and k are non-negative integers and $h < k$. Note that each derivative is with respect to x_t ; this sum is agnostic about variation in x following an initial shock.

This expression can be solved by substituting from Equation 4 in the main text and simplifying the resulting geometric series. Just as the instantaneous effect of x on y is slightly different from all other period-specific effects, as the shock to x has not yet worked its way through the system of variables, so too is $t = 0$ a special case for cumulative effects. For $h > 0$, the expression for the cumulative effect is given by

$$\sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} = \frac{(\beta_0 + \beta_4 z_t + \beta_6 z_{t-1})(\alpha_1^h - \alpha_1^{k+1}) + (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t)(\alpha_1^{h-1} - \alpha_1^k)}{1 - \alpha_1}, \quad (\text{SM.29})$$

but for $h = 0$, it is simply

$$\sum_{j=0}^k \frac{\partial y_{t+j}}{\partial x_t} = \frac{(\beta_0 + \beta_4 z_t + \beta_6 z_{t-1})(1 - \alpha_1^{k+1}) + (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t)(1 - \alpha_1^k)}{1 - \alpha_1}. \quad (\text{SM.30})$$

For the calculation of the LRM, we allow k to approach infinity and observe:

$$\sum_{j=0}^{\infty} \frac{\partial y_{t+j}}{\partial x_t} = \frac{\beta_0 + \beta_1 + \beta_4 z_t + \beta_5 z_{t+1} + \beta_6 z_{t-1} + \beta_7 z_t}{1 - \alpha_1} \quad (\text{SM.31})$$

Of course, the maximum possible effect would be observed over the longest time horizon, which is why the LRM receives so much attention. By no means is the LRM the only quantity available to scholars for describing long-run relationships. There are many time intervals, short and long, that can be constructed to study the behavior of the dynamic system. These could be calculated by substituting some other value k into Equation SM.30.

Finite-period cumulative effects are useful for answering many questions scholars care about which the LRM cannot. Most notable are questions about fixed time periods such as an electoral cycle, parliamentary session, or European Union presidency. Using the LRM to describe such relationships is unreliable because the difference between the cumulative effect from zero to k and from zero to infinity is often non-trivial.¹ It is also increasing in α_1 , so the LRM is less useful for making inferences about finite periods where y is more strongly autoregressive. In other words, the longer it takes for effects to accumulate or dissipate over time, the worse the end result describes what's going on in the interim. Even very large infinite-period relationships are not necessarily good approximations of finite-period relationships.

Further, finite-period cumulative effects are arguably more interesting and substantively important than the LRM. For instance, the question “What is the effect of an economic shock on public opinion over the length of a presidential term?” cannot be meaningfully answered with the LRM. Even if it is possible to calculate how economic downturn affects presidential popularity from now until forever, the many elections in between may make this exercise more of a curiosity than an empirically relevant prediction, particularly where data constraints bind. Examining cumulative effects over various time windows can ameliorate the problem of extrapolating from short time series, and help scholars make more

¹This difference can be found by subtracting Equation SM.30 from Equation SM.31, and equals $\frac{\alpha_1^k (\alpha_1 [\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}] + \beta_1 + \beta_5 z_{t+1} + \beta_7 z_t)}{1 - \alpha_1}$.

precise statements about the substantive significance of their findings.

4.1.3 Threshold effects

The last quantity of interest that can be recovered from Equation 2 are threshold effects, defined as the number of periods following a shock to x required for the total change in y to reach some theoretically-relevant value λ . These are closely related to the lag lengths discussed in De Boef and Keele (2008), which define λ as a proportion (δ) of the LRM. Thus threshold effects may answer “How many years would it take for democratization to raise per capita income by \$100?” while lag lengths may answer “How many years after democratization does 80% of total change in income accrue?”

Threshold effects and lag lengths are calculated by setting Equation SM.30 equal to some λ and solving for the period k . How to interpret them depends on whether the cumulative effect of x on y is monotonically increasing over time. If it is, then only thresholds $|\lambda| < |\text{LRM}|$ produce sensible answers, since a threshold bigger than the total effect will never be reached. If it is not, then we choose a threshold $|\text{LRM}| < |\lambda|$ and solve for the period k during which the cumulative effect drops below λ in absolute value. Since thresholds are difficult to compute in closed analytic form, we can simply calculate cumulative effects over increasing k and visually inspect the results to find the period where the predicted values cross λ . Evaluating cumulative effects over a variety of periods is an important part of interpreting such models in general, so thresholds can be studied with trivial added effort.

Scholars commonly discuss these effects only in the context of median and mean lag lengths, which describe the period at which half ($\delta = .5$) and all ($\delta = 1$) of the LRM has accumulated, respectively.² However, thresholds of interest need not be set to one of only two values, but rather can be used to examine any quantile of the LRM. Nor must threshold

²De Boef and Keele (2008, 192) argue that scholars should always report these “forgotten quantities.” Yet the lag length for $\delta = 1$ is typically either nonsensical or uninteresting. If the cumulative effect of x on y is monotonic, then $\delta = 1$ is undefined, since the LRM is always greater (in absolute value) than any finite-period cumulative effect to some decimal place. If it is non-monotonic, then $\delta = 1$ simply yields $t = 0$, as in the Casillas et al. (2011) example above. The inferential value of this quantity is therefore limited.

effects be expressed as a proportion of the LRM at all. Absolute levels of change in y may be of greater interest, especially for policy applications. As with the other quantities described above, a variety of threshold effects can be used to convey information about the specific dynamic system under examination.

4.2 Formulae for conditional relationships in the ADL framework

4.2.1 Period-specific effects

Recall that the general conditional model (assuming a lag order of 1 for x and z) is given by

$$\begin{aligned} y_t = & \alpha_0 + \alpha_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 z_t + \beta_3 z_{t-1} \\ & + \beta_4 x_t z_t + \beta_5 x_{t-1} z_t + \beta_6 x_t z_{t-1} + \beta_7 x_{t-1} z_{t-1} + \epsilon_t. \end{aligned} \quad (\text{SM.32})$$

which is Equation 3 in the main text. Period-specific effects are defined as $\frac{\partial y_{t+j}}{\partial x_t} \forall j \in \mathbb{Z}$. We first examine the case where $j = 0$, which is just Equation SM.32. Differentiating with respect to x_t yields

$$\frac{\partial y_t}{\partial x_t} = \beta_0 + \beta_4 z_t + \beta_6 z_{t-1}.$$

Moving to $j = 1$, the model becomes

$$\begin{aligned} y_{t+1} = & \alpha_0 + \alpha_1 y_t + \beta_0 x_{t+1} + \beta_1 x_t + \beta_2 z_{t+1} + \beta_3 z_t \\ & + \beta_4 x_{t+1} z_{t+1} + \beta_5 x_t z_{t+1} + \beta_6 x_{t+1} z_t + \beta_7 x_t z_t + \epsilon_{t+1}, \end{aligned}$$

into which we can substitute the right side of Equation SM.32 for y_t . This yields

$$\begin{aligned}
 y_{t+1} = & \alpha_0 + \alpha_1 (\alpha_0 + \alpha_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 z_t + \beta_3 \\
 & + z_{t-1} + \beta_4 x_t z_t + \beta_5 x_{t-1} z_t + \beta_6 x_t z_{t-1} + \beta_7 x_{t-1} z_{t-1} + \epsilon_t) \\
 & + \beta_0 x_{t+1} + \beta_1 x_t + \beta_2 z_{t+1} + \beta_3 z_t \\
 & + \beta_4 x_{t+1} z_{t+1} + \beta_5 x_t z_{t+1} + \beta_6 x_{t+1} z_t + \beta_7 x_t z_t + \epsilon_{t+1}.
 \end{aligned}$$

Differentiating yields

$$\frac{\partial y_{t+1}}{\partial x_t} = \alpha_1 (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) + \beta_1 + \beta_5 z_{t+1} + \beta_7 z_t,$$

which is the period-specific effect for $j = 1$. Following the same process for $j = 2$ gives $\frac{\partial y_{t+2}}{\partial x_t} = \alpha_1^2 (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) + \alpha_1 (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t)$. This pattern continues for all $j \in \mathbb{Z}_{>0}$, with each set of terms in parentheses multiplied by α_1 to increasingly higher powers. Finally, note that for any $j < 0$, no substitutions can be made such that an x_t term appears among the covariates, making each derivative zero. Thus, for negative j , all period-specific effects are nil. This gives the general expression for period-specific effects, the equations for which are shown in Equation SM.27.

4.2.2 Cumulative effects

Cumulative effects are sums of period-specific effects. Defined most generally as

$$\sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} \equiv \frac{\partial y_{t+h}}{\partial x_t} + \frac{\partial y_{t+h+1}}{\partial x_t} + \dots + \frac{\partial y_{t+k}}{\partial x_t},$$

we can substitute such that

$$\begin{aligned}
\sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} &= \alpha_1^h (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) + \alpha_1^{h-1} (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t) \\
&\quad + \alpha_1^{h+1} (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) + \alpha_1^h (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t) \\
&\quad \vdots \\
&\quad + \alpha_1^k (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) + \alpha_1^{k-1} (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t).
\end{aligned}$$

Group terms and rewrite:

$$\begin{aligned}
\sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} &= \left(\alpha_1^h + \alpha_1^{h+1} + \dots + \alpha_1^k \right) (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) \\
&\quad + \left(\alpha_1^{h-1} + \alpha_1^h + \dots + \alpha_1^{k-1} \right) (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t). \tag{SM.33}
\end{aligned}$$

These are two geometric sums which can be solved simultaneously. Multiply both through by α_1 and then subtract the result from Equation SM.33, giving

$$\begin{aligned}
(1 - \alpha_1) \sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} &= \left(\alpha_1^h - \alpha_1^{k+1} \right) (\beta_0 + \beta_4 z_t + \beta_6 z_{t-1}) \\
&\quad + \left(\alpha_1^{h-1} - \alpha_1^k \right) (\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t).
\end{aligned}$$

Dividing through by $(1 - \alpha_1)$ gives the expression for $0 < h < k$ (Equation SM.31). Note, however, that when $h = 0$, this expression raises $\beta_1 + \beta_5 z_{t+1} + \beta_7 z_t$ to a negative power. This is not correct, and the instantaneous effect does not include any of these terms. Thus $h = 0$ presents a special case, for which repeating the same approach to derivation with $h = 0$ yields the general expression in Equation SM.30.

4.2.3 Threshold effects

For ease of presentation, define

$$\mathbf{a} = \beta_0 + \beta_4 z_t + \beta_6 z_{t-1}$$

$$\mathbf{b} = \beta_1 + \beta_5 z_{t+1} + \beta_7 z_t.$$

Also define an arbitrary threshold λ , and solve for k in Equation SM.30 where $h = 0$. This yields

$$k = \left\lceil \frac{\log \left(\frac{\mathbf{a} + \mathbf{b} - (1 - \alpha_1) \lambda}{\alpha_1 \mathbf{a} + \mathbf{b}} \right)}{\log \alpha_1} \right\rceil,$$

where the ceiling function indicates rounding up since k is a discrete period in time. Lag lengths are identically derived, where $\lambda \equiv \delta \Omega$, $\delta \in [0, 1)$, and Ω is the LRM. As discussed, both quantities are more easily found inductively than solved for.

4.3 Formulae for conditional relationships in the ECM framework

Recall that the general model with a conditional relationship is given in ADL form in Equation SM.32 and ECM form in Equation SM.1. All of the quantities described in Section 4.1 can be recovered from either specification; Table SM.1 provides the variable translations to move between models. These can be used to rewrite the general equations for quantities of interest from the ECM, provided here.

Table SM. 1: Variable transformations between the general ADL and ECM in Equations SM.32 and SM.1

ADL	ECM
$\alpha_0 = \gamma_0$	$\alpha_0 = \gamma_0$
$\alpha_1 = \gamma_1 + 1$	$\alpha_1 - 1 = \gamma_1$
$\beta_0 = \theta_0$	$\beta_0 = \theta_0$
$\beta_1 = \theta_1 - \theta_0$	$\beta_0 + \beta_1 = \theta_1$
$\beta_2 = \theta_2$	$\beta_2 = \theta_2$
$\beta_3 = \theta_3 - \theta_2$	$\beta_2 + \beta_3 = \theta_3$
$\beta_4 = \theta_4 + \theta_5 + \theta_6$	$\beta_4 - \beta_6 = \theta_4$
$\beta_5 = -(\theta_4 + \theta_6)$	$\beta_4 - \beta_5 = \theta_5$
$\beta_6 = -(\theta_5 + \theta_6)$	$-(\beta_4 + \beta_5 + \beta_6) = \theta_6$
$\beta_7 = \theta_6 + \theta_7$	$\beta_4 + \beta_5 + \beta_6 + \beta_7 = \theta_7$

4.3.1 Period-specific effects

Equation SM.27 becomes

$$\frac{\partial y_{t+j}}{\partial x_t} = \begin{cases} 0 & \text{for } j \in \mathbb{Z}_{<0}, \\ \theta_0 + \theta_4 z_t + (\theta_5 + \theta_6) \Delta z_t & \text{for } j = 0, \\ (\gamma_1 + 1)^j [\theta_0 + \theta_4 z_t + (\theta_5 + \theta_6) \Delta z_t] + \\ \quad (\gamma_1 + 1)^{j-1} [\theta_1 - \theta_0 - \theta_4 z_{t+1} - \theta_6 \Delta z_{t+1} + \theta_7 z_t] & \text{for } j \in \mathbb{Z}_{>0}. \end{cases} \quad (\text{SM.34})$$

4.3.2 Cumulative effects

$$\sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} = \frac{(\theta_0 + \theta_4 z_t + \theta_5 \Delta z_t + \theta_6 \Delta z_t) \left([\gamma_1 + 1]^h - [\gamma_1 + 1]^{k+1} \right)}{-\gamma_1} + \frac{(\theta_1 - \theta_0 - \theta_4 z_{t+1} - \theta_6 \Delta z_{t+1} + \theta_7 z_t) \left([\gamma_1 + 1]^{h-1} - [\gamma_1 + 1]^k \right)}{-\gamma_1}$$

for $h, k \in \mathbb{Z}_{>0}$ such that $h < k$, and

$$\sum_{j=h}^k \frac{\partial y_{t+j}}{\partial x_t} = \frac{(\theta_0 + \theta_4 z_t + \theta_5 \Delta z_t + \theta_6 \Delta z_t) \left(1 - [\gamma_1 + 1]^{k+1}\right)}{-\gamma_1} + \frac{(\theta_1 - \theta_0 - \theta_4 z_{t+1} - \theta_6 \Delta z_{t+1} + \theta_7 z_t) \left(1 - [\gamma_1 + 1]^k\right)}{-\gamma_1}$$

for $h = 0$. Also note that the LRM is given by

$$\sum_{j=0}^{\infty} \frac{\partial y_{t+j}}{\partial x_t} = \frac{\theta_1 + \theta_7 z_t + (\theta_5 + \theta_6) \Delta z_t - (\theta_4 + \theta_6) \Delta z_{t+1}}{-\gamma_1}. \quad (\text{SM.35})$$

4.3.3 Threshold effects

First define

$$\mathbf{a} = \theta_0 + \theta_4 z_t + (\theta_5 + \theta_6) \Delta z_t$$

$$\mathbf{b} = \theta_1 - \theta_0 - \theta_4 z_{t+1} - \theta_6 \Delta z_{t+1} + \theta_7 z_t.$$

for ease of presentation. Also define an arbitrary threshold λ , and solve for k from the cumulative effect where $h = 0$. This yields

$$k = \left\lceil \frac{\log \left(\frac{\mathbf{a} + \mathbf{b} - (1 - \alpha_1) \lambda}{\alpha_1 \mathbf{a} + \mathbf{b}} \right)}{\log \alpha_1} \right\rceil,$$

where the ceiling function indicates rounding up since k is a discrete period in time. Lag lengths are identically derived, where $\lambda \equiv \delta \Omega$, $\delta \in [0, 1)$, and Ω is the LRM. As discussed, both quantities are more easily found inductively than solved for.

4.4 Estimating uncertainty with parametric bootstrapping

It can be difficult to calculate these effects while also defining measures of uncertainty. Building on King et al. (2000), we can describe the parametric bootstrap for a maximum likelihood framework:

1. Estimate the model, e.g. Equation SM.32, from which we wish to draw inferences.
2. Store the point estimates ($\hat{\beta}$) and variance-covariance matrix ($\mathbb{C}(\hat{\beta})$).
3. Take M draws from the multivariate normal $\mathcal{N}_{MV}(\hat{\beta}, \mathbb{C}(\hat{\beta}))$ and save the output.
4. Define S scenarios of interest to study. For instance, we may hold all variables at their central tendency but vary one along its interquartile range by S increments. With dynamic models, these scenarios need to satisfy the constraint $\Delta y_t = y_t - y_{t-1}$ (and similarly for x and z) to be empirically relevant.
5. For each $s \in S$ and $m \in M$, calculate quantities of interest.
6. For each $s \in S$, take quantiles over the M draws, e.g. the .025th and .975th, to approximate 95% confidence intervals.

This procedure yields statements of uncertainty for every quantity of interest for every scenario $s \in S$. Example code for this procedure is provided in the replication archive.

4.5 Empirical example

Kono and Montinola (2013) provide an example. The authors investigate the effect of official development assistance (ODA) on military spending, conditional on regime type. Ignoring control variables, their specification is the ECM

$$\begin{aligned} \Delta y_t = & \gamma_0 + \gamma_1 y_{t-1} + \theta_0 \Delta x_t + \theta_1 x_{t-1} + \theta_2 \Delta z_t + \theta_3 z_{t-1} + \\ & \theta_4 \Delta x_t z_{t-1} + \theta_7 x_{t-1} z_{t-1} + \epsilon_t, \end{aligned}$$

where y is military expenditure as a proportion of gross national income (GNI), x is ODA as a proportion of GNI, and z is regime type (variously measured). This specification restricts $\theta_5 = \theta_6 = 0$, implying that there is no interaction that involves change in regime type, consistent with the authors' theoretical expectations. Yet deriving the LRM for this model yields $\frac{\theta_1 + \theta_7 z_t - \theta_4 \Delta z_t}{-\gamma_1}$, indicating that the effect of a change in ODA is conditional on change in, not just level of, democracy. Setting aside regime change could therefore lead to incomplete or incorrect conclusions about the dynamic system. The only way to protect inferences from such threats is to derive quantities of interest *for each specific model*.

Period-specific, cumulative, and threshold effects can all be reported with statements of uncertainty such as confidence intervals. Perhaps the most easily implemented method of calculating uncertainty is parametric bootstrapping, which requires fitting a model and then sampling from the distribution of the estimated parameters King et al. (2000). More specifically, we can store the point estimates and variance-covariance matrix from the fitted model ($\hat{\beta}$ and $\mathbb{C}[\hat{\beta}]$ from Equation SM.32), and then take M draws from the multivariate normal $\mathcal{N}_{MV}(\hat{\beta}, \mathbb{C}[\hat{\beta}])$. Any quantity of interest defined above can then be computed for each draw $m \in M$. The estimated effect is given by the mean across all M draws, and confidence intervals can be computed from the quantiles (e.g., the 95% confidence interval is provided by the 2.5% and 97.5% quantiles). Example R code for this procedure is provided in the replication materials.

This approach can be used alongside standard methods for estimating uncertainty, such as the “delta method,” which occasionally have the limitation of not being easily extended to complex models nor to all of these quantities of interest. In these cases, parametric bootstrapping resolves this problem. By giving uncertainty estimates for all of the quantities we want to study, parametric bootstrapping allows scholars to make much richer inferences about dynamic systems.

5 Further Monte Carlo results

5.1 Investigating simulation experimental conditions

Our Monte Carlo experiments investigate multiple conditions. Recall, we examine $3^4 = 81$ cases, varying the time series length $n \in \{50, 100, 500\}$ and the autoregressive parameters $\rho_x, \rho_z, \rho_y \in \{0.1, 0.5, 0.9\}$. In the main text, we do not delineate between results from these parameters. Readers might wonder if they contribute meaningfully to the variation in the figures. To answer, we reproduce Figure 1 from the main text in Figures SM. 1, 2, and 3. We then reproduce Figure 2 from the main text in Figures SM. 4, 5, and 6. The broad patterns from the main text hold across all Figures.

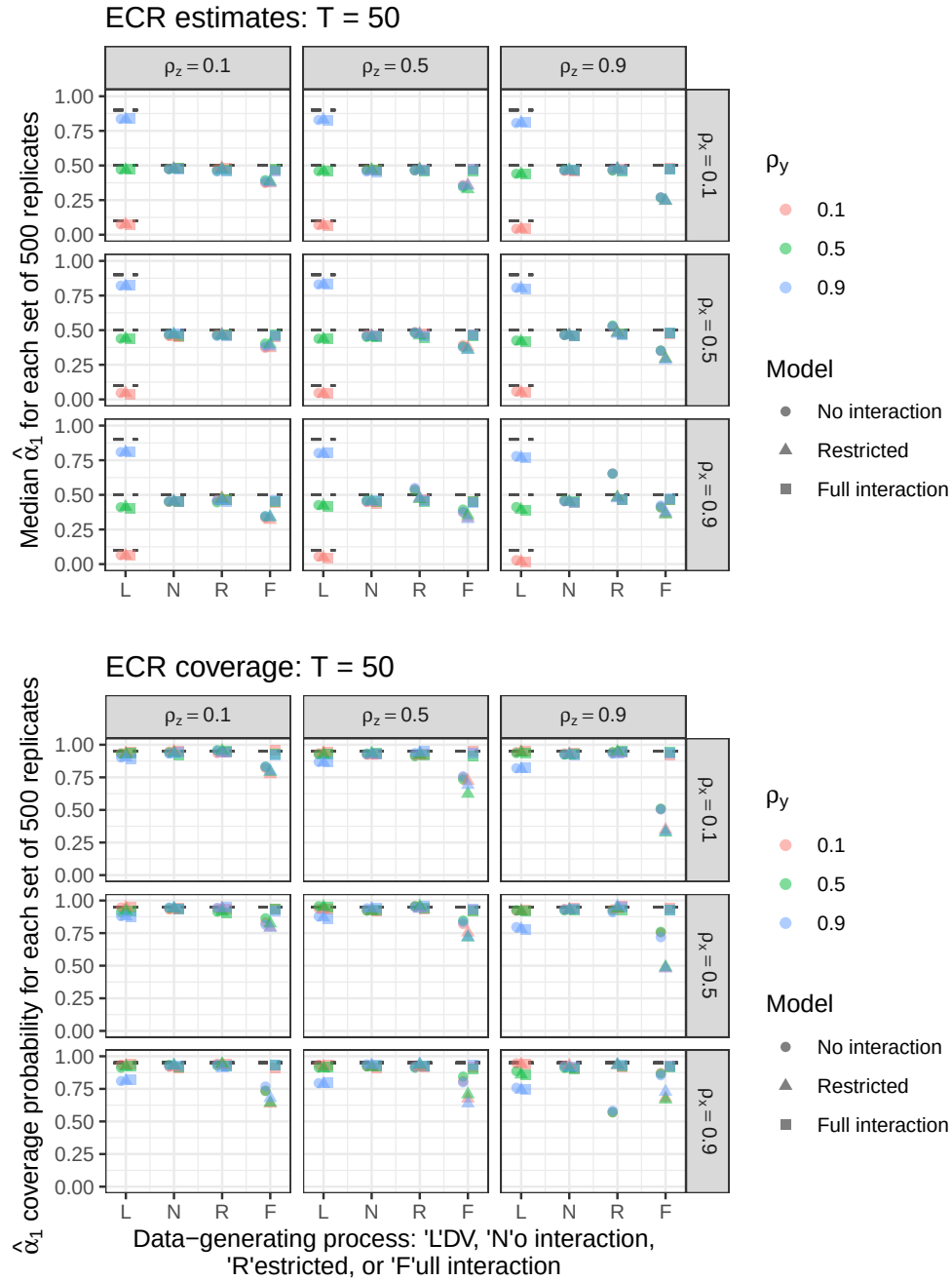


Figure SM. 1: Monte Carlo simulation results for the error correction rate, α_1 . Triangles represent estimates from the model without an interaction, squares are from a restricted model, and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value. Estimates are featured by possible levels of ρ ; $T = 50$.

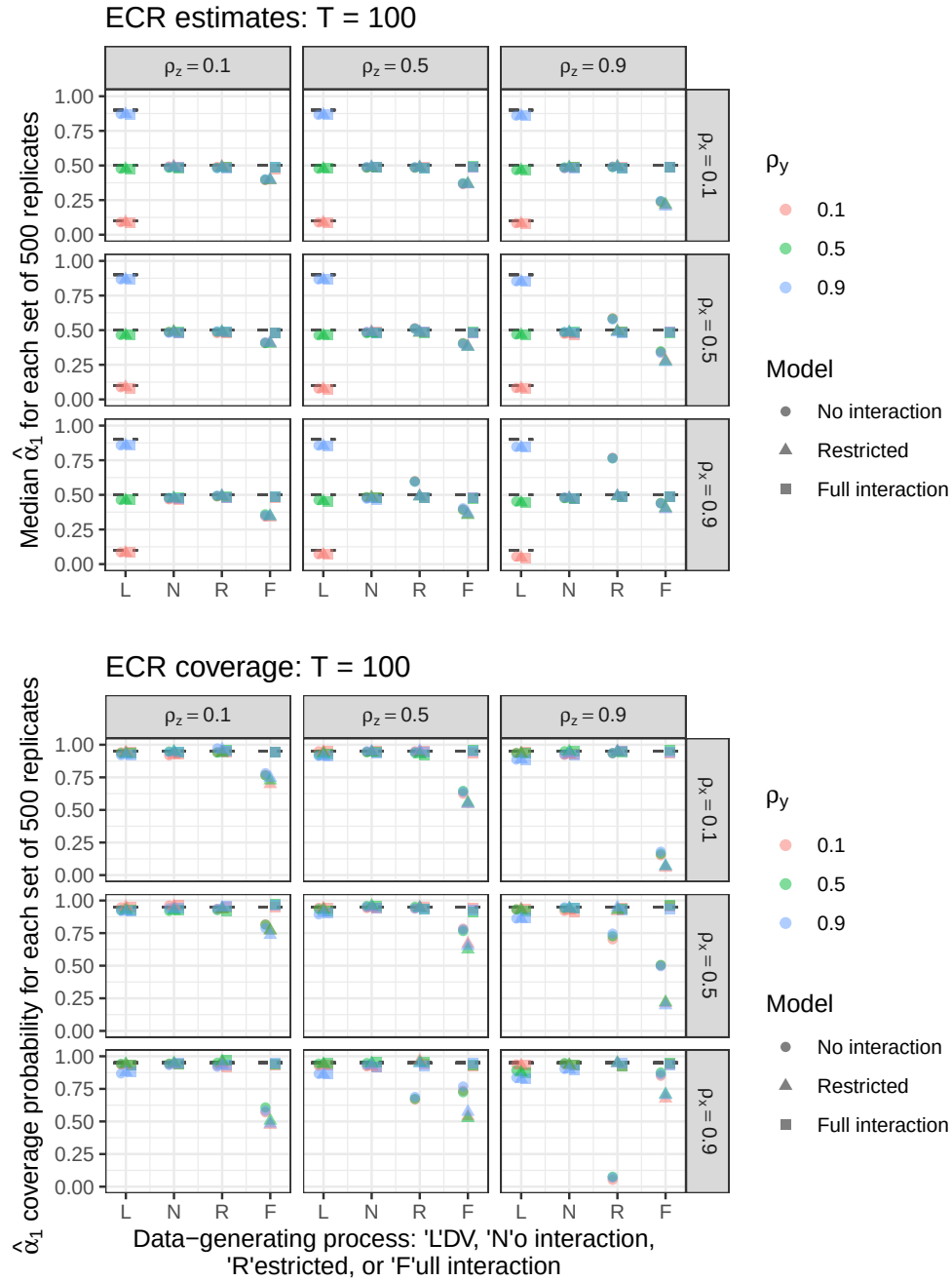


Figure SM. 2: Monte Carlo simulation results for the error correction rate, α_1 . Triangles represent estimates from the model without an interaction, squares are from a restricted model, and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value. Estimates are featured by possible levels of ρ ; $T = 100$.

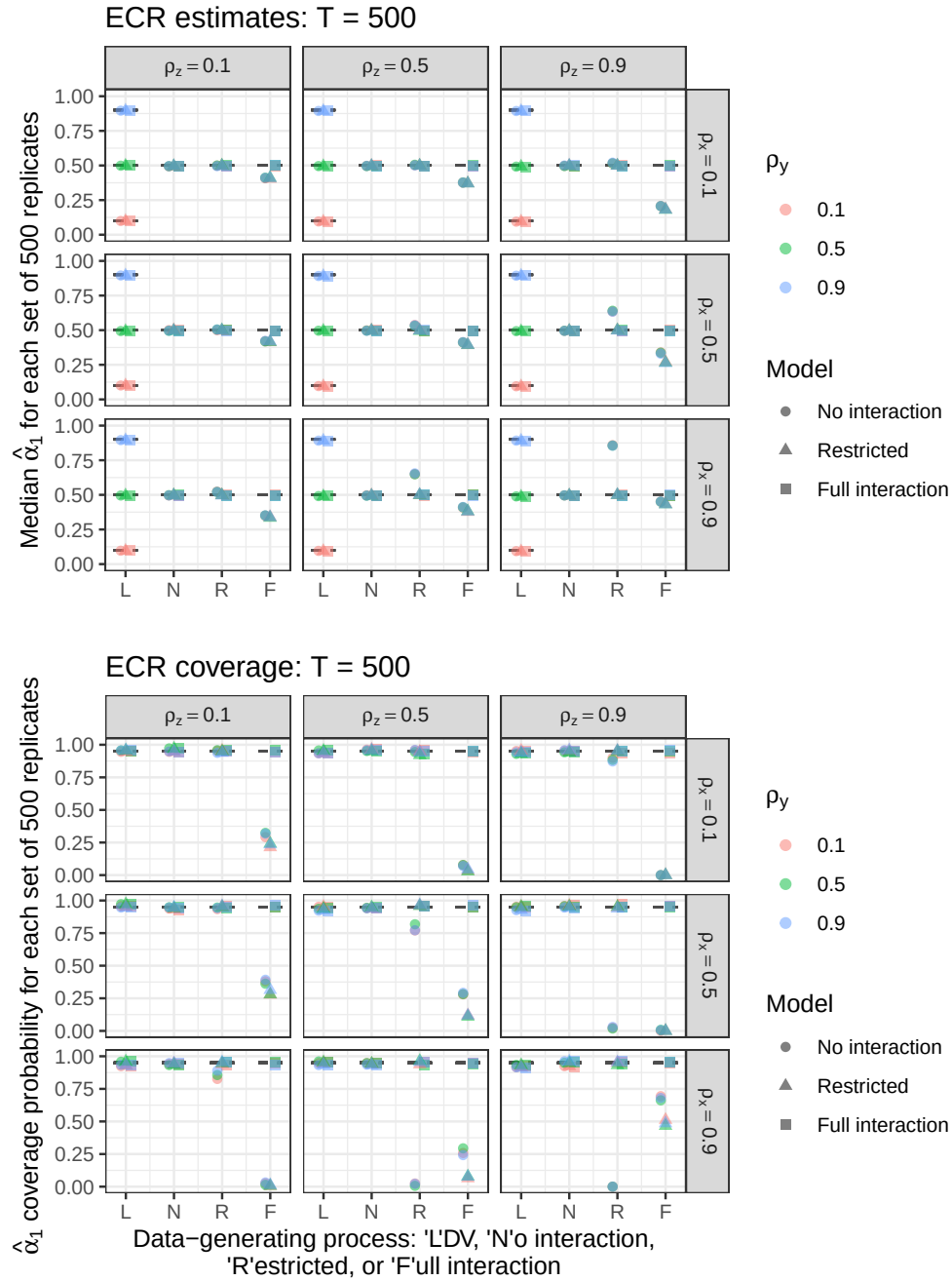


Figure SM. 3: Monte Carlo simulation results for the error correction rate, α_1 . Triangles represent estimates from the model without an interaction, squares are from a restricted model, and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value. Estimates are featured by possible levels of ρ ; $T = 500$.

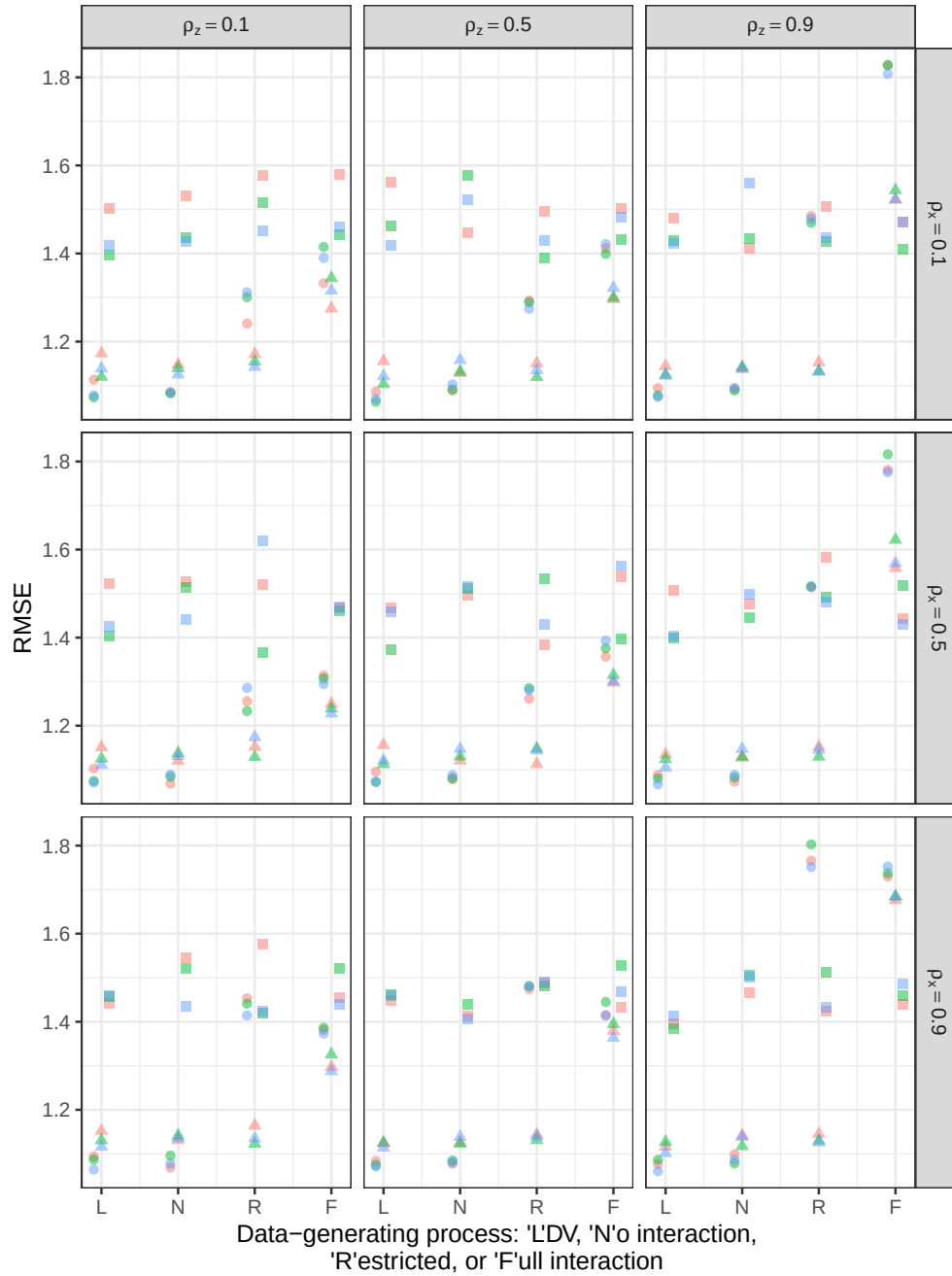


Figure SM. 4: Monte Carlo simulation results for RMSE from 5-fold cross-validation. Triangles represent RMSE from the model without an interaction, squares are from a restricted model, and circles are from the general model. Each point represents the median RMSE from a 20% random sample of each set of 500 replications under each experimental condition. Estimates are featured by possible levels of ρ ; $T = 50$.

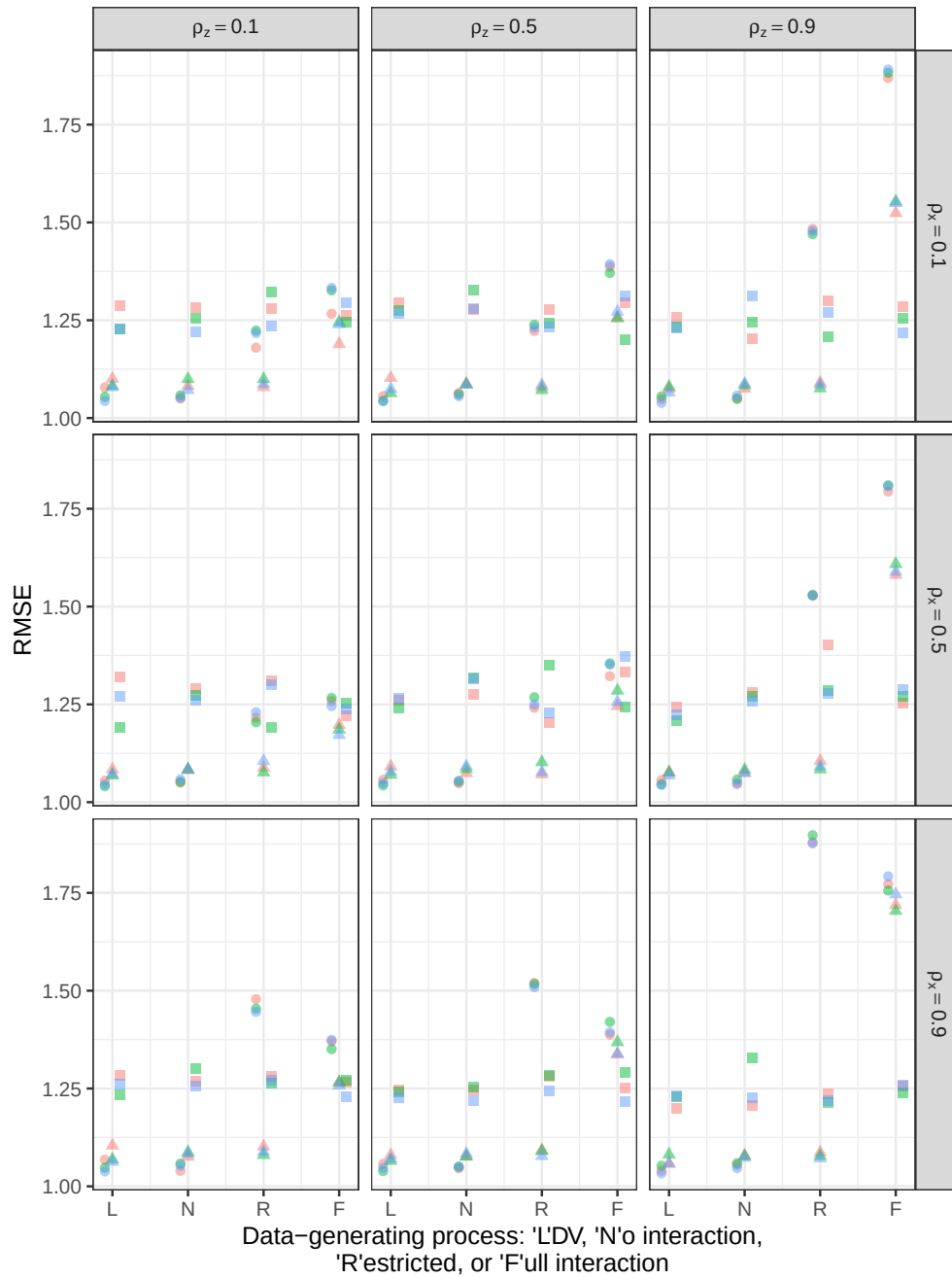


Figure SM. 5: Monte Carlo simulation results for RMSE from 5-fold cross-validation. Triangles represent RMSE from the model without an interaction, squares are from a restricted model, and circles are from the general model. Each point represents the median RMSE from a 20% random sample of each set of 500 replications under each experimental condition. Estimates are featured by possible levels of ρ ; $T = 100$.

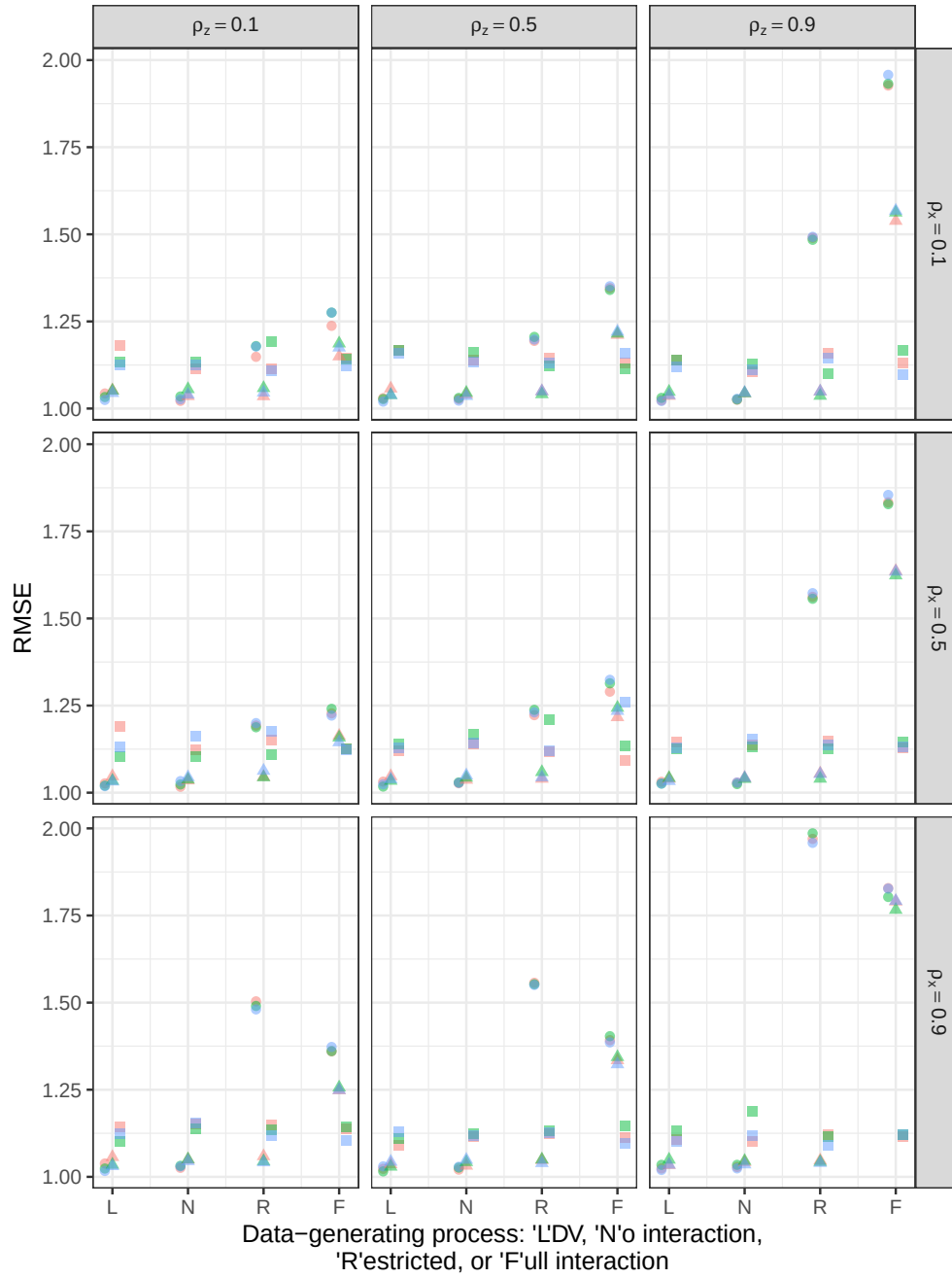


Figure SM. 6: Monte Carlo simulation results for RMSE from 5-fold cross-validation. Triangles represent RMSE from the model without an interaction, squares are from a restricted model, and circles are from the general model. Each point represents the median RMSE from a 20% random sample of each set of 500 replications under each experimental condition. Estimates are featured by possible levels of ρ ; $T = 500$.

5.2 Other parameters

In this section we provide further Monte Carlo results. Each figure plots median estimates in the top panel and coverage rates in the bottom panel, analogous to Figure 1 for the parameters of interest not discussed in the text. Like with the error correction rate, we find that the general model produces better estimates of *all* parameters than does a model with an invalid restriction.

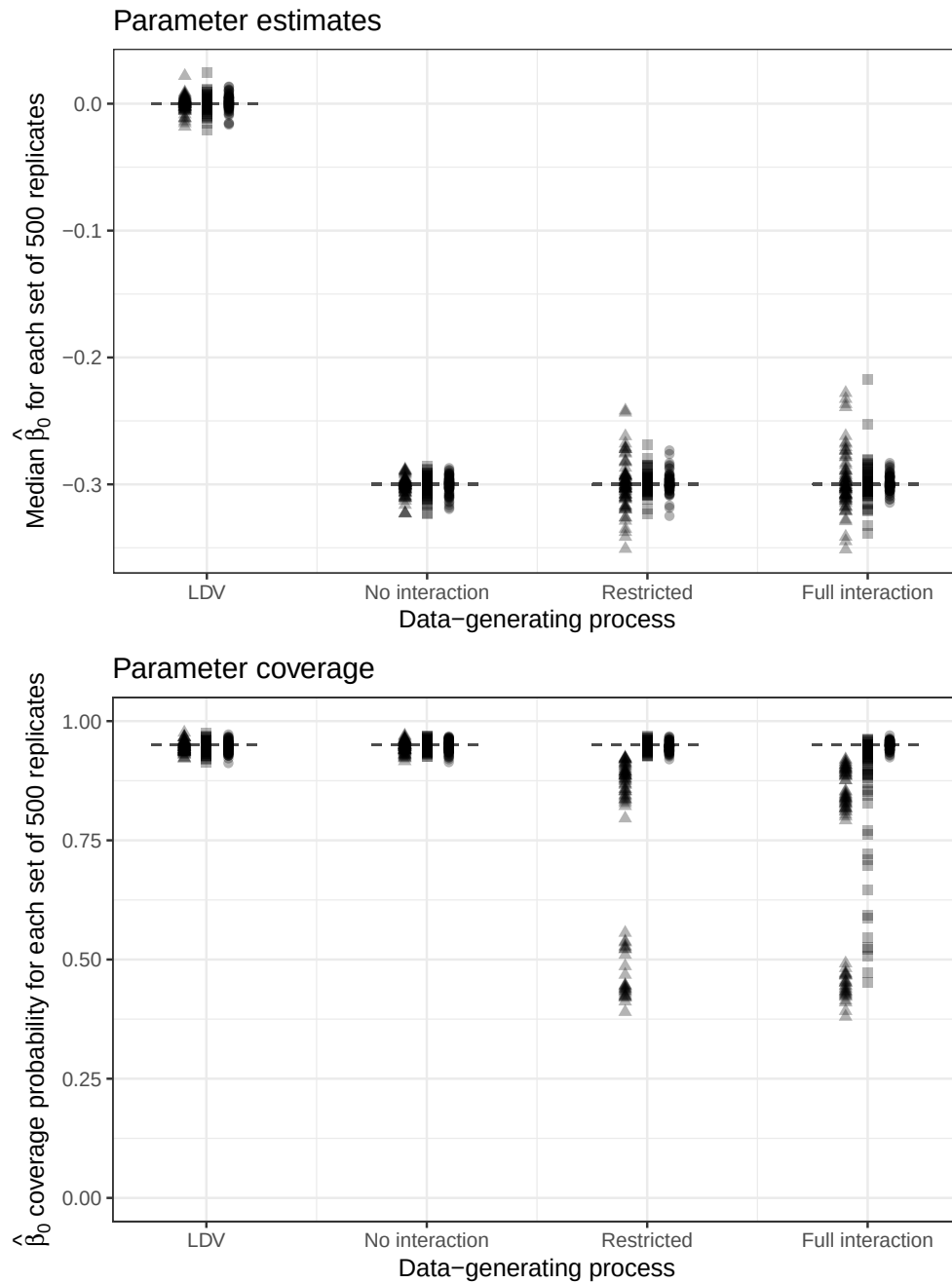


Figure SM. 7: Monte Carlo simulation results for β_0 . Triangles represent estimates from the model without an interaction, squares are from a restricted model, and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

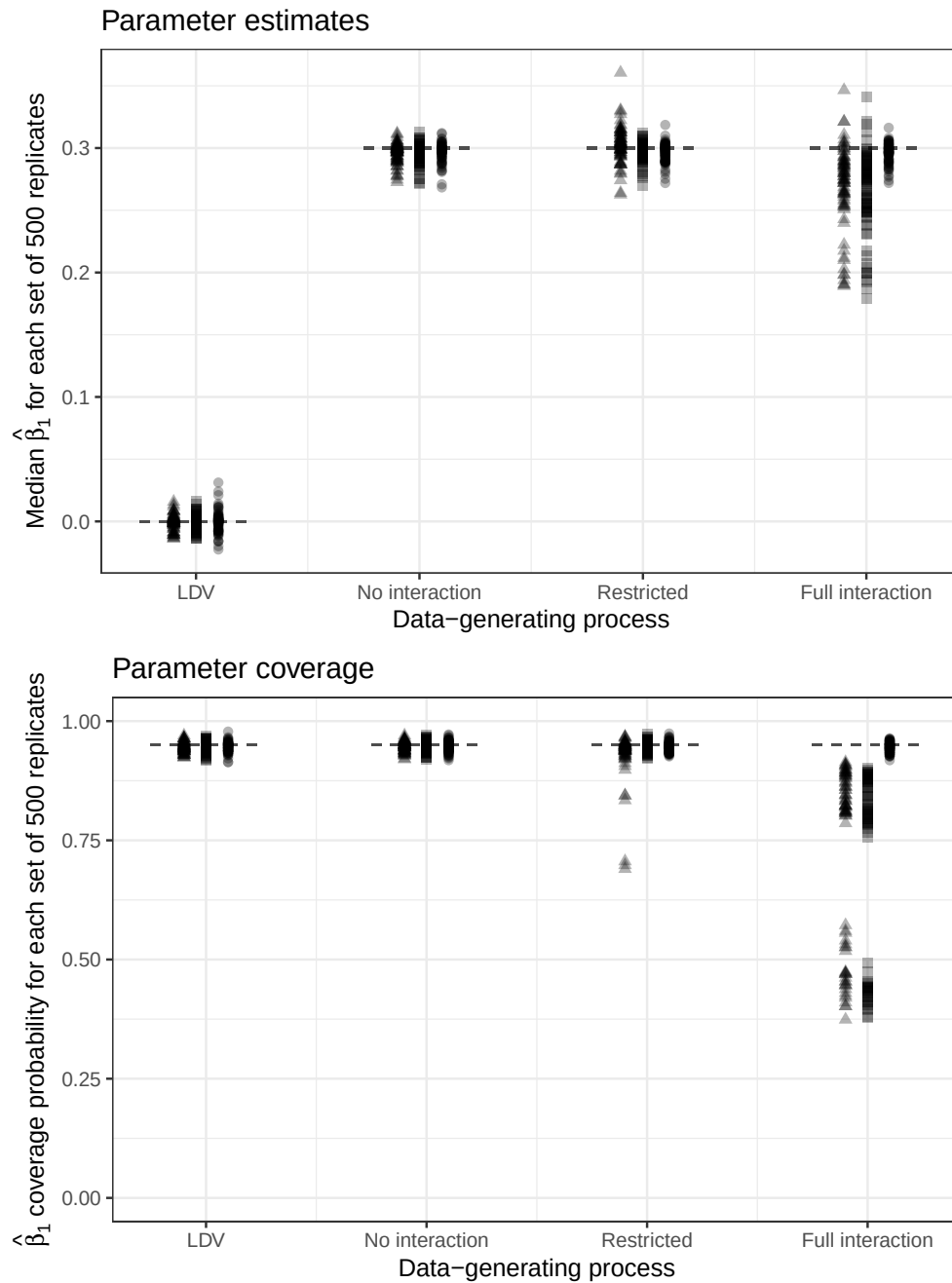


Figure SM. 8: Monte Carlo simulation results for β_1 . Triangles represent estimates from the model without an interaction, squares are from a restricted model, and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

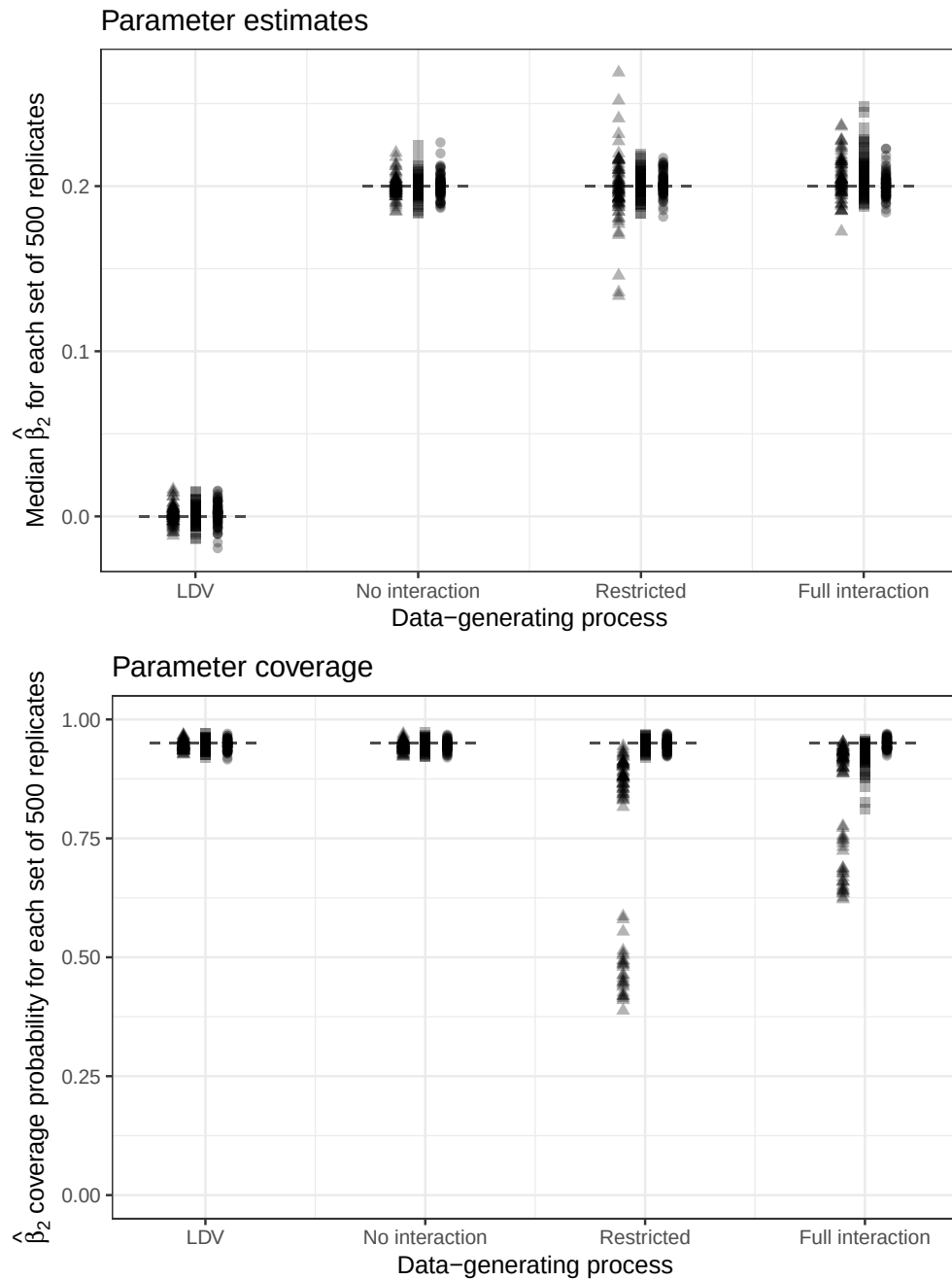


Figure SM. 9: Monte Carlo simulation results for β_2 . Triangles represent estimates from the model without an interaction, squares are from a restricted model, and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

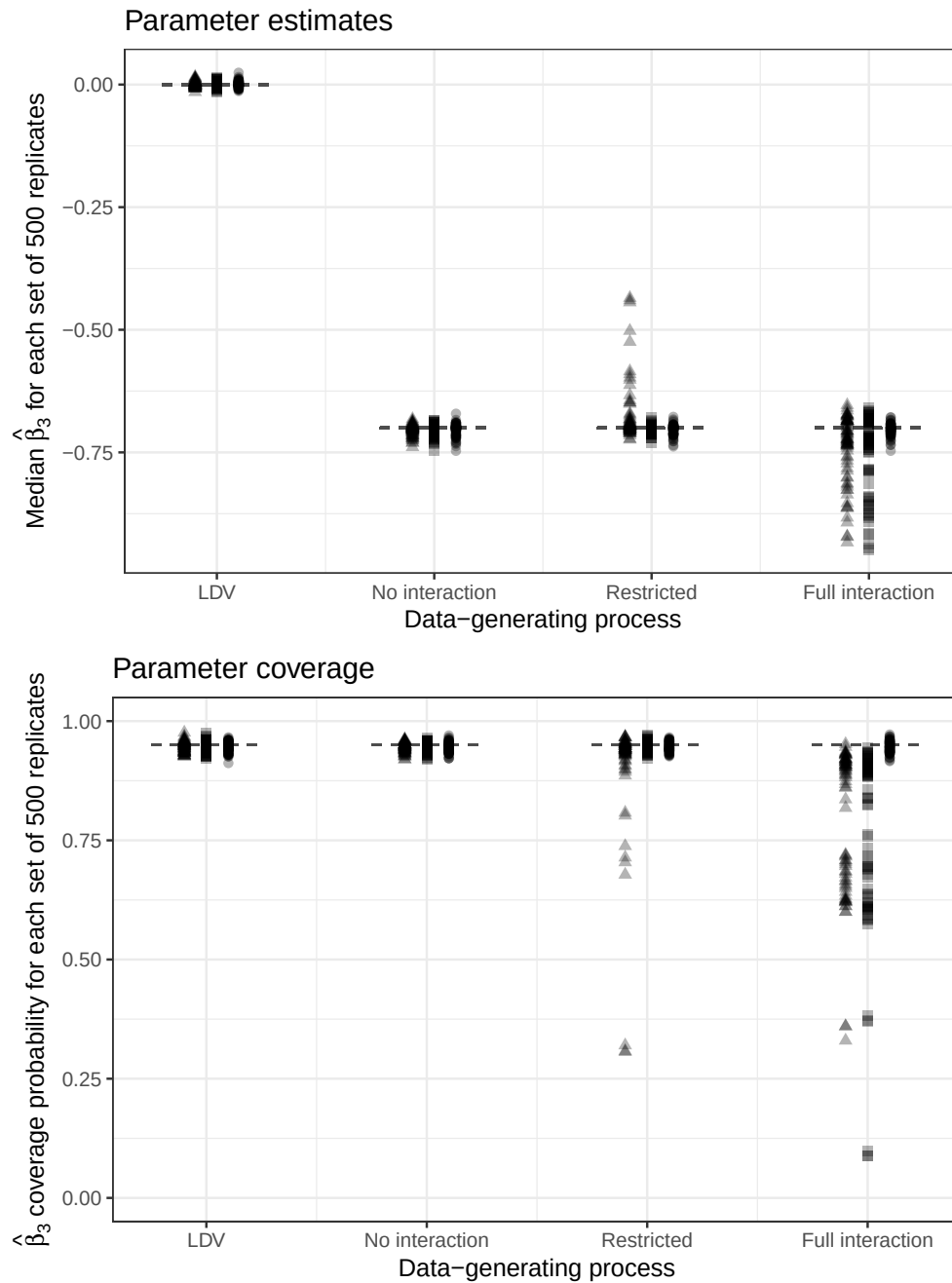


Figure SM. 10: Monte Carlo simulation results for β_3 . Triangles represent estimates from the model without an interaction, squares are from a restricted model, and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

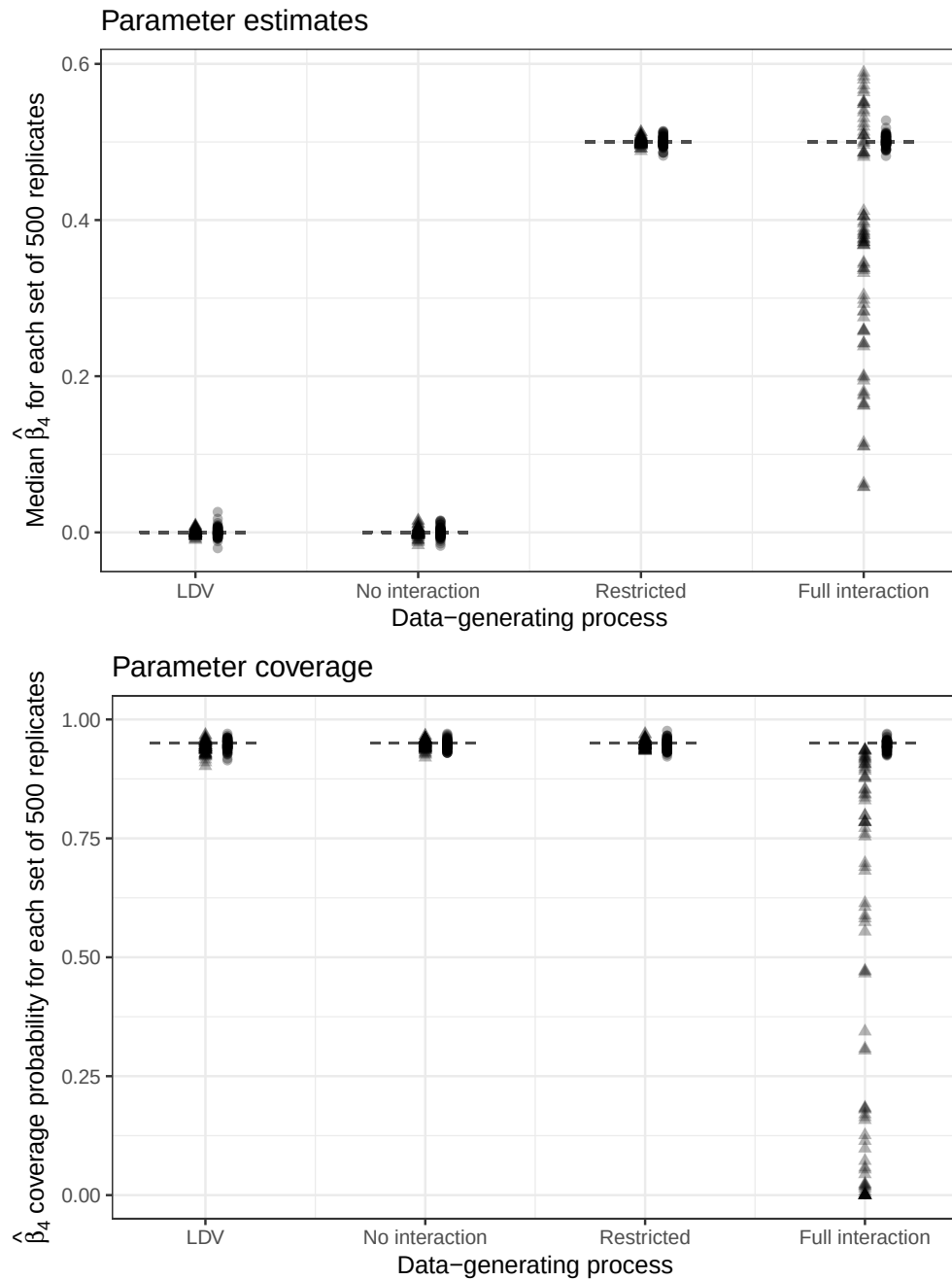


Figure SM. 11: Monte Carlo simulation results for β_4 . Squares represent estimates from a restricted model and circles are from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

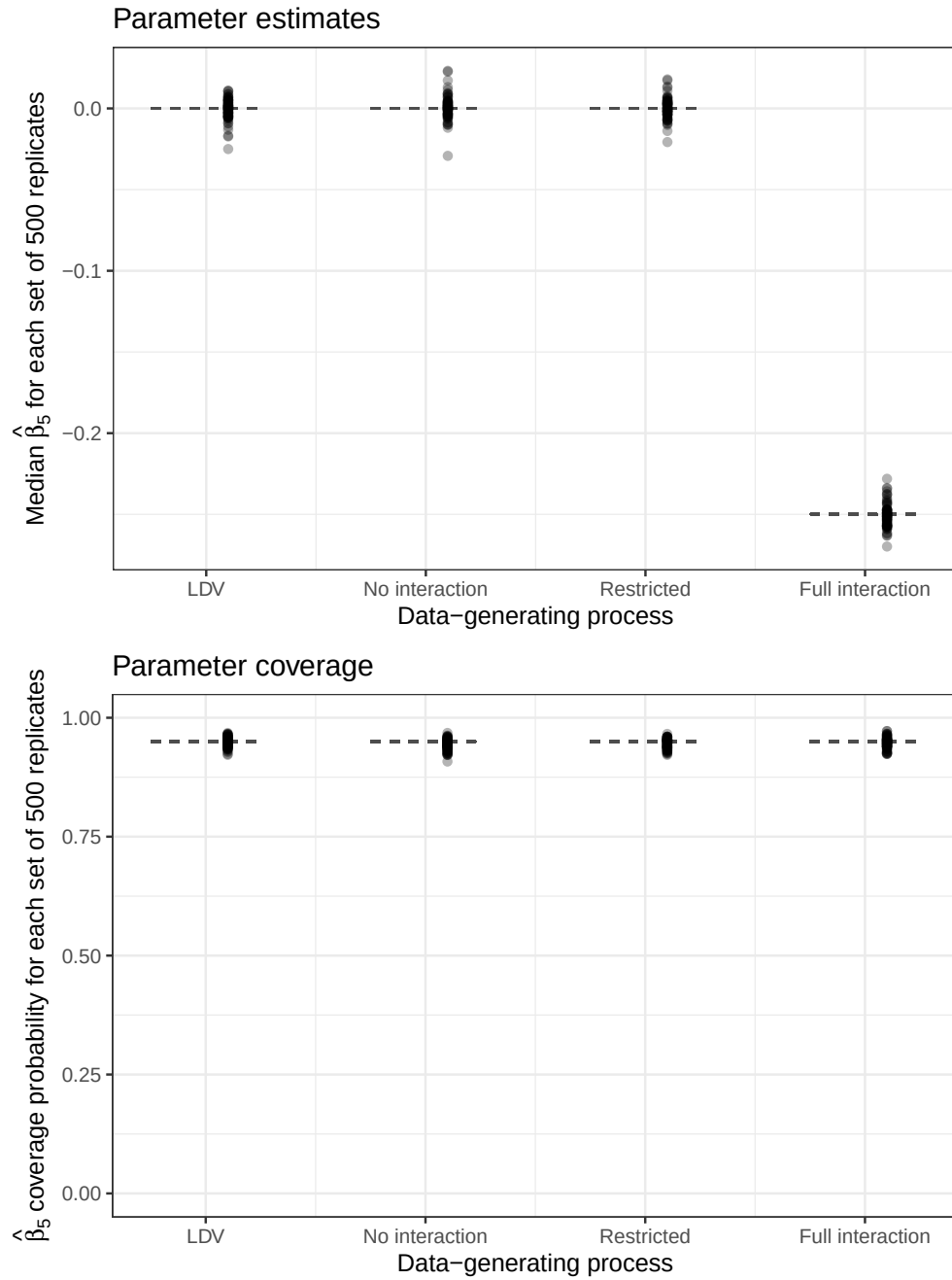


Figure SM. 12: Monte Carlo simulation results for β_5 . Circles represent estimates from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

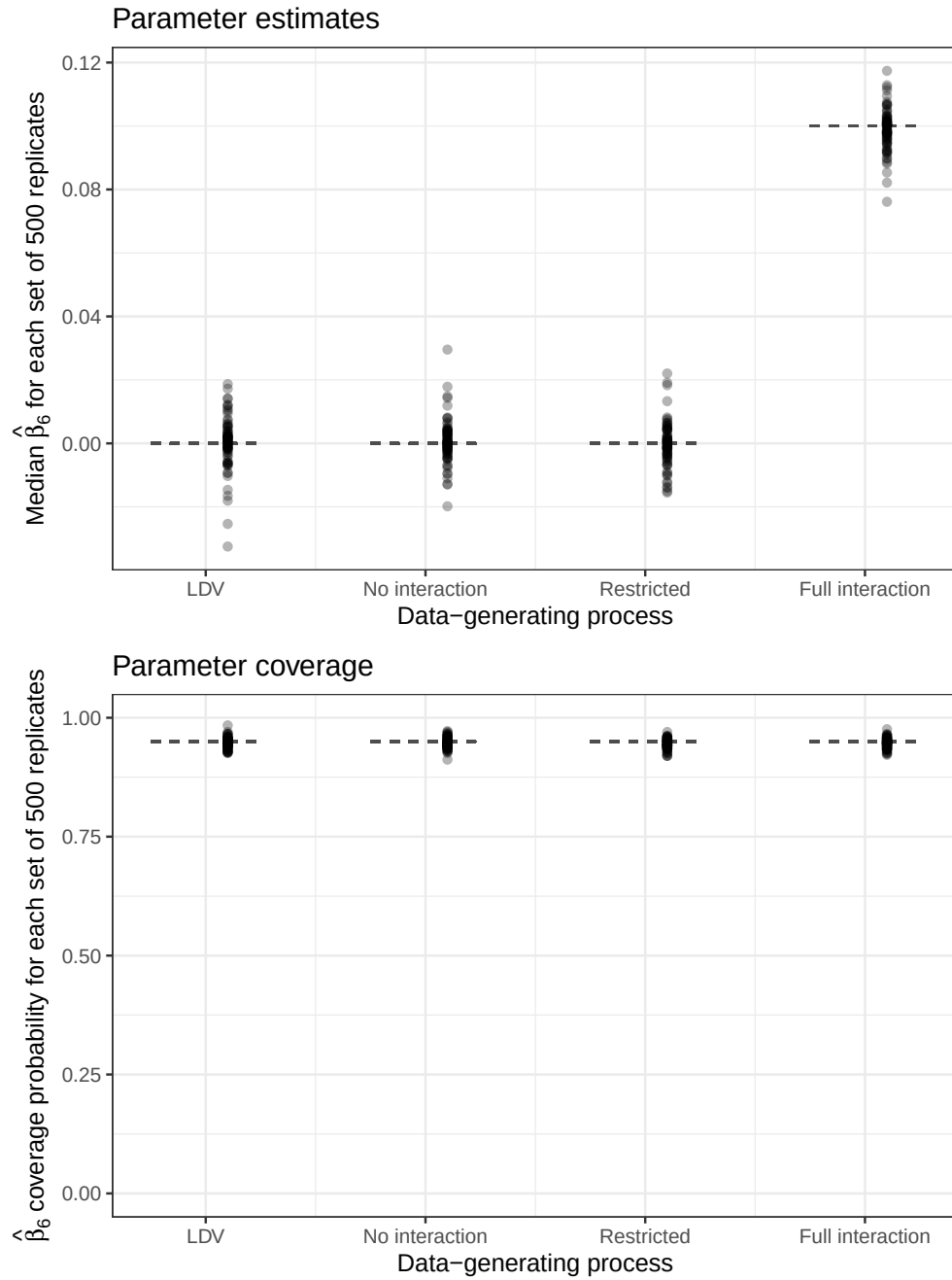


Figure SM. 13: Monte Carlo simulation results for β_6 . Circles represent estimates from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

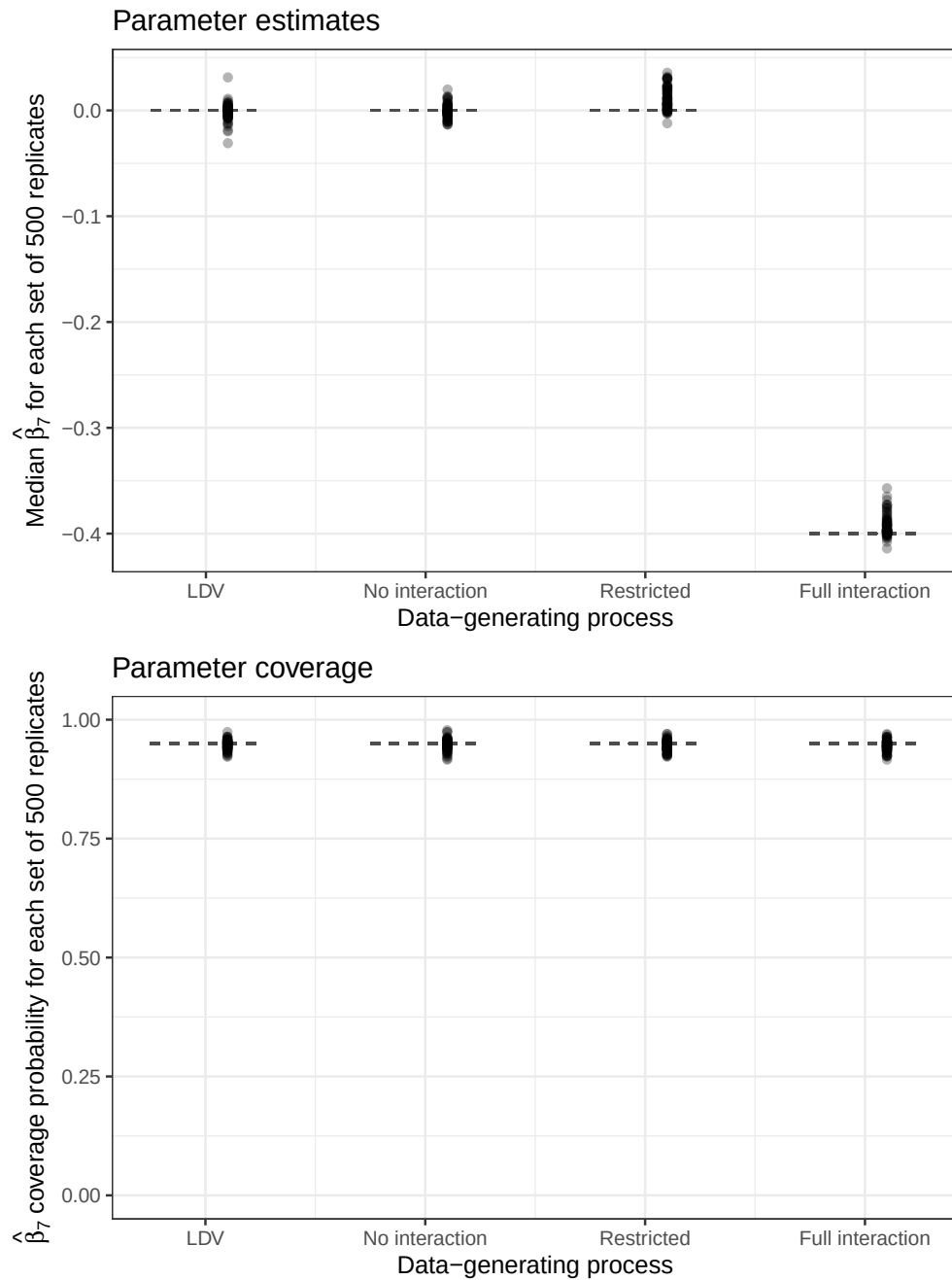


Figure SM. 14: Monte Carlo simulation results for β_7 . Circles represent estimates from the general model. Median estimates are plotted against the true values (dashed lines) in the top panel. The bottom panel plots the proportion of 500 replicates under each condition for which 95% confidence intervals contain the true value.

5.3 Statistical power

An additional is the extent to which the fully interactive model has the statistical power to detect relationships, given the number of parameters it is estimating. Figure SM.15 sheds light on this question. The y-axis shows the mean proportion of 500 replicates in which the relevant fully interactive parameter (β_0 to β_7) was estimated to be statistically significant; the boxplot shows the distribution of these means over our simulation conditions. Our expectations are

1. For the “LDV” (red) DGP, no parameters should be statistically significant
2. For the “No interaction” (green) DGP, only β_0 to β_3 should be statistically significant
3. For the “Restricted” (teal) DGP, β_0 to β_4 should be statistically significant
4. For the “Full interaction” (purple) DGP, all β s should be statistically significant

This is broadly what we find. Of particular note are the teal boxplots: when the DGP is the restricted interaction, these proportions are still centered at 0.05 for β_5 , β_6 , and β_7 , suggesting the fully interactive model is not finding spurious interactions. As expected, this performance is somewhat worse when $T = 50$, such that β_4 is only statistically significant about 75% of the time (as opposed to 95%). Given an even arbitrarily longer period ($T = 500$), the estimates for both β_4 and β_7 behave as expected. The only relatively poor performance comes from the cross-lag parameters β_5 and β_6 . Even with a long time series ($T = 500$), the cross-lag parameter is only statistically significant about 55% of the time. This is suggestive evidence that the fully interactive model excels at estimating the direct interactions (β_4 and β_7) but finds difficulty in parsing the cross-lag component. As such, practitioners should take real caution in interpreting any single component of these models independently.

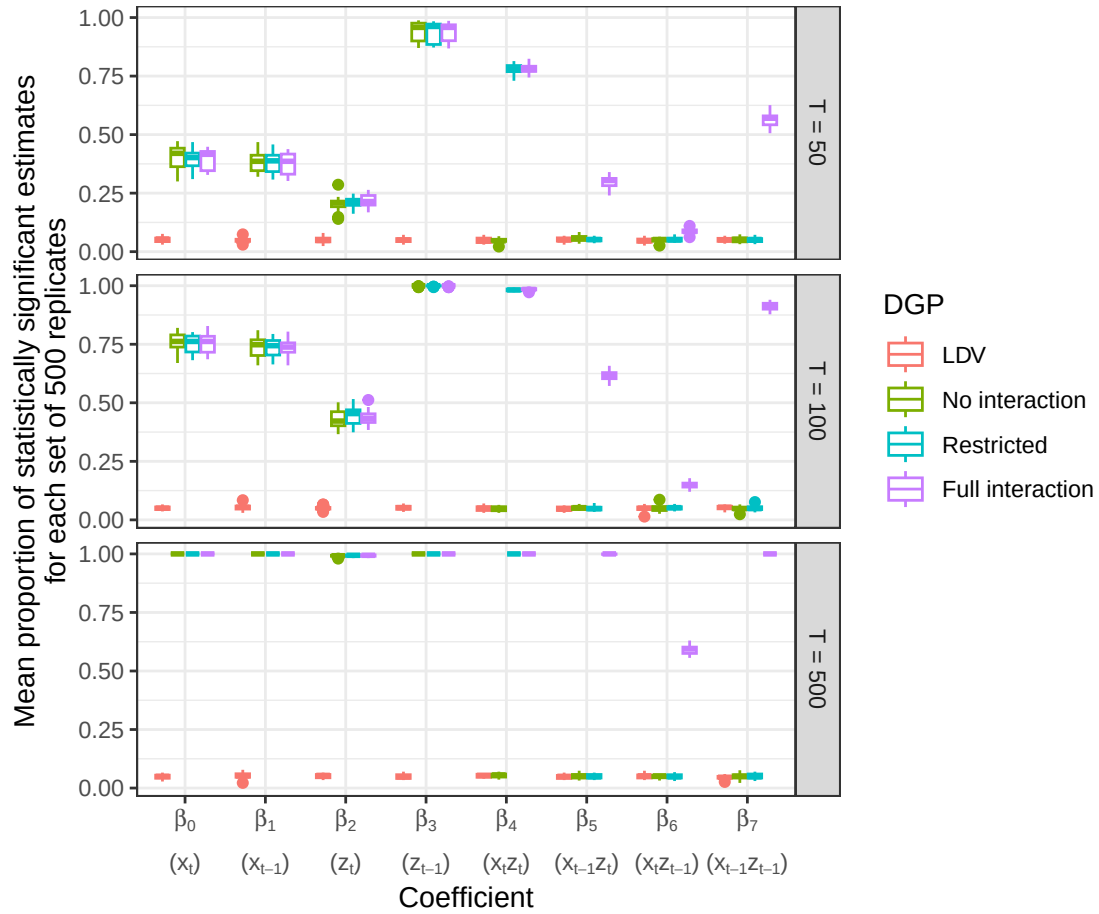


Figure SM. 15: Monte Carlo simulation results for statistical power. Each boxplot is the proportion of 500 replicates under each condition for which the relevant parameter is estimated to be statistically significant in the fully interactive model, given the DGP.

6 Cavari (2019) tabular results

Here we present the tabular results for the Cavari replication models estimated in the main paper.

Table SM. 2: Replication of Cavari (2019): comparing the restricted and general models

	Cavari	General	Restricted
Performance: economy	0.10 (0.12)	0.64* (0.14)	0.70* (0.14)
Performance: foreign affairs	0.18 (0.12)	0.44* (0.11)	0.41 (0.11)
MIP: economy	-0.78* (0.16)	-	-
MIP: foreign affairs	-0.69* (0.20)	-	-
Performance: economy $\times$ MIP: economy	0.02* (0.003)	-	-
Performance: foreign affairs $\times$ MIP: foreign affairs	0.01* (0.004)	-	-
Δ MIP: economy	-	-0.31 (0.29)	-0.32 (0.22)
Δ MIP: foreign affairs	-	0.11 (0.51)	0.22 (0.48)
Performance: economy _{$t-1$}	0.10 (0.12)	-0.41* (0.15)	-0.45* (0.14)
Performance: foreign affairs _{$t-1$}	0.18 (0.12)	-0.19 (0.11)	-0.20 (0.11)
Δ MIP: economy _{$t-1$}	-	0.29 (0.30)	0.18 (0.29)
Δ MIP: foreign affairs _{$t-1$}	-	0.45 (0.38)	0.48 (0.37)
Performance: economy $\times$ Δ MIP: economy	-	-0.003 (0.02)	-0.01 (0.02)
Performance: foreign affairs $\times$ Δ MIP: foreign affairs	-	-0.02* (0.01)	-0.02* (0.01)
Performance: economy _{$t-1$} $\times$ Δ MIP: economy _{$t-1$}	-	-0.03 (0.02)	-0.04 (0.02)
Performance: foreign affairs _{$t-1$} $\times$ Δ MIP: foreign affairs _{$t-1$}	-	-0.03 (0.03)	-0.03 (0.03)
Performance: economy $\times$ Δ MIP: economy _{$t-1$}	-	0.03 (0.02)	0.03 (0.02)
Performance: economy _{$t-1$} $\times$ Δ MIP: economy	-	0.01 (0.03)	0.02 (0.02)
Performance: foreign affairs $\times$ Δ MIP: foreign affairs _{$t-1$}	-	0.02 (0.03)	0.02 (0.03)
Performance: foreign affairs _{$t-1$} $\times$ Δ MIP: foreign affairs	-	0.02 (0.02)	0.02 (0.01)
Approval rating _{$t-1$}	0.12 (0.08)	0.61* (0.09)	0.61* (0.10)
Macro-partisanship (in-party)	0.42* (0.15)	0.42* (0.21)	0.13 (0.13)
Macro-partisanship (out-party)	0.27* (0.13)	-0.02 (0.17)	-
Macro-partisanship (in-party) _{$t-1$}	-	-0.34 (0.20)	-
Macro-partisanship (out-party) _{$t-1$}	-	0.28 (0.16)	0.19 (0.12)
Unemployment rate	0.45 (0.60)	-	-
Δ Unemployment rate	-	-0.89 (1.46)	-
Δ Unemployment rate _{$t-1$}	-	-0.51 (1.15)	-
Divided government	0.98 (0.92)	0.43 (0.90)	-
Election year	-0.28 (0.69)	-0.90 (0.77)	-
Honeymoon	-0.82 (1.61)	-1.98 (1.73)	-
Observations	133	133	133
R ²	0.91	0.90	0.90

* $p < .05$. The dependent variable is presidential approval. Omitted from table are the constant and presidential fixed effects.

7 Further replications

7.1 Blaydes and Kayser (2011)

7.1.1 Data and estimation

In this section we demonstrate how the modeling and inferential framework developed above can improve our understanding of important dynamic relationships.³ Blaydes and Kayser (2011, “B & K”) investigate a central problem in political economy: whether regime type conditions pro-poor growth in developing countries. Many formal models of regime transition rely on the assumption that democracies redistribute better than autocracies (Acemoglu and Robinson, 2006; Boix, 2003), yet the evidence for this assumption is mixed (Houle, 2009).

B & K make two innovations in addressing this question. First, they theorize that clientelism may be driving inequality reduction in democracies, generating a new empirical implication: if the gains from growth are shared through non-programmatic redistribution, then scholars should expect inequality reduction primarily in “hybrid” (or semi-democratic) regimes, where provision of clientelist goods is most common. Second, B & K introduce average daily caloric consumption as a proxy for inequality. Biological limits on the number of calories an individual can consume mean that the rich are not able to capture economic growth as it is transmitted through the production of basic foodstuffs, unlike increases in the value of capital or physical assets. Moreover, since the middle and upper classes consume near-ideal calories, any increase in the national average is likely to be driven by gains among the poor. This interpretation of consumption habits is substantiated by comparing calories with inequality figures generated elsewhere (Deininger and Squire, 1996). Thus B & K hypothesize that the effect of growth on calories consumed is larger in hybrid regimes and democracies than in autocracies.

³This application uses a panel, which is less than ideal for illustration purposes. However, we wanted to demonstrate how the ideas could extend to this context.

Data on caloric consumption are gathered annually by the Food and Agriculture Organization, covering nearly every country in the world over the period 1961-2003. The figures are calculated by taking the initial stock of food, adding what was produced or imported, and subtracting what was exported, estimated to have been wasted, or left in the national stock at end of year. The result provides an estimate of annual consumption, which is then normalized by population. For the independent variables, B & K bin country-years by Polity IV regime type (Marshall et al., 2011), creating three categories: autocracies (≤ -7), hybrid regimes (-6 to 6), and democracies (≥ 7). Growth is calculated by first-differencing gross domestic product (GDP) per capita, which is expressed in hundreds of constant 2000 US dollars. As this literature is concerned primarily with developing countries, Blaydes and Kayser (2011) drop country-years with a GDP per capita of \$10,000 or higher. We follow their coding decisions to ensure comparability.⁴

We first replicate B & K's preferred model, given by

$$\Delta y_t = \gamma_0 + \gamma_1 y_{t-1} + \theta_0 \Delta x_t + \theta_1 x_{t-1} + \theta_3 \mathbf{z}_{t-1} + \theta_4 \Delta x_t \mathbf{z}_{t-1} + \theta_8 \mathbf{w} + \epsilon_t, \quad (\text{SM.36})$$

where y is average per capita daily caloric consumption, x is GDP per capita, $\mathbf{z}$ is a vector of regime type indicators, $\mathbf{w}$ is a vector of country fixed effects,⁵ and ϵ is idiosyncratic error. Since countries rarely change regime types, B & K do not include multiple periods of $\mathbf{z}$. As a result, only one period of regime type is interacted with growth, forcing $\theta_2 = \theta_5 = \theta_6 = \theta_7 = 0$. In addition to this restricted specification, we also estimate the general model in Equation SM.1 but with the country dummies retained.

Before estimation, we pretest the data to ensure that the ECM is appropriate. We examine the order of integration for each series, as well as the interaction between growth and

⁴B & K omit five observations with outlier values for growth, which we also drop. 69 further observations drop out due to missingness; to facilitate model comparison, we use the remaining sample to estimate all specifications. We also follow the authors in providing heteroskedastic-robust standard errors without clustering.

⁵Adding fixed effects to a model that includes the lagged dependent variable likely induces (Nickell, 1981) bias. We include them here to maintain comparability across studies, since B & K use them to account for cross-sectional heterogeneity. Note, however, that our results are robust to estimating the general model without country fixed effects.

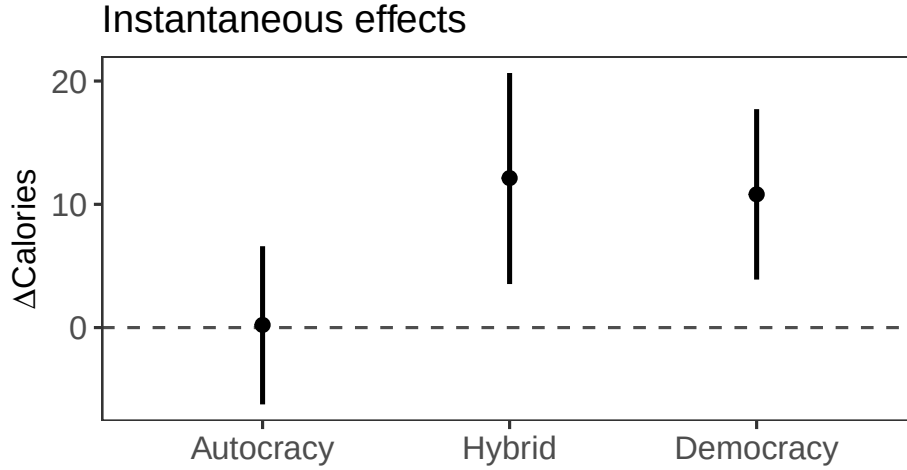


Figure SM. 16: Short-run relationships between growth and inequality by regime type. Each point represents the predicted instantaneous effect on average daily per capita calories consumed of a \$100 increase in per capita GDP, conditional on regime type, from the full model. 95% confidence intervals constructed from quantiles of 5,000 samples from $\mathcal{N}_{MV}(\hat{\theta}, \mathbb{C}(\hat{\theta}))$ are plotted as lines.

regime type, using ADF, KPSS, and Ljung-Box tests, and visually inspect Autocorrelation Function (ACF) and partial ACF (PACF) plots. These diagnostics are complicated by the structure of the data (an unbalanced panel with short time series), but the preponderance of evidence indicates that the interaction is a unit root process. Thus, the interaction is integrated, and—by Proposition 2—cointegrated with the other variables, which we confirm using the Engle-Granger and Johansen cointegration tests. As a whole, these results suggest that the ECM and ADL are appropriate models for these data, and OLS estimates will be consistent.

7.1.2 Results

Results from these models are presented in Table SM.3. Column 2 replicates the headline results from Blaydes and Kayser (2011) and column 3 gives the estimates for the general model.⁶ The results appear generally consistent across models, with only the coefficient on

⁶Estimates for the B & K model reported here differ from the published results because of the slightly smaller sample, but nothing changes in substantive or statistical significance.

Table SM. 3: Replication of Blaydes and Kayser (2011): comparing the restricted and general models

	B & K model	General model
Calories _{t-1}	-0.08* (0.01)	-0.08* (0.01)
ΔGDP_t	3.51* (1.43)	0.26 (3.28)
GDP _{t-1}	0.31 (0.30)	0.49 (0.43)
$\Delta\text{Regime type}_t$		-6.30 (3.27)
Hybrid _{t-1}	12.95* (3.88)	18.45* (4.84)
Democracy _{t-1}	9.67 (5.29)	11.08 (7.30)
$\Delta\text{GDP}_t \times \text{Hybrid}_{t-1}$	9.96* (3.15)	11.84* (3.16)
$\Delta\text{GDP}_t \times \text{Democracy}_{t-1}$	10.50* (2.50)	10.54* (2.51)
$\text{GDP}_{t-1} \times \Delta\text{Regime type}_t$		0.13 (0.17)
$\Delta\text{GDP}_t \times \Delta\text{Regime type}_t$		2.94 (2.84)
$\text{GDP}_{t-1} \times \text{Hybrid}_{t-1}$		-0.63* (0.29)
$\text{GDP}_{t-1} \times \text{Democracy}_{t-1}$		-0.25 (0.32)
Observations	3,264	3,264
R ²	0.10	0.11
RMSE	77.83	77.32

* $p < .05$. The dependent variable is $\Delta\text{Calories}$. Both models include 112 country fixed effects, with heteroskedastic-robust standard errors. RMSE is calculated from out-of-sample predictions using 5-fold cross-validation.

economic growth changing significance. Yet they produce very different predicted effects. B & K report that GDP growth of \$100 increases caloric consumption for all countries, with the effect significantly larger in hybrid regimes and democracies. In contrast, Figure SM.16 plots instantaneous effects for the general model, holding each country's regime type constant; here we cannot distinguish instantaneous effects across regime types, despite the large and significant coefficients on the interaction terms. All that can be said is that there

is some evidence that growth is associated with greater caloric consumption among the poor in hybrid regimes and democracies.

More differences between the models' predictions emerge in the dynamic effects. The top and bottom panels of Figure SM.17 plot period-specific and cumulative effects, respectively. Examining the period-specific effects, we have essentially no information except that, following an initial shock, the effect of growth on caloric consumption is negative in hybrid regimes. However, the trend is very similar in democracies, so it is difficult to draw sharp distinctions among regime types. Turning now to cumulative effects, we see that the instantaneous and long-run effects overlap for each regime. None of the LRMs—6 calories in autocracies, -2 in hybrid regimes, and 3 in democracies—are statistically distinguishable from zero. These results contrast sharply with those of B & K, who report LRMs of 13, 170, and 187, respectively. Additionally, the LRMs overlap across all regimes, meaning that it is not possible to infer whether the effect of growth on caloric consumption differs across regime types. To formally test for an effect of regime type, we would need to test for the statistical difference of these LRMs; for our purposes, though, we simply note that the conditional relationship is much less defined than B & K indicate.

Threshold effects provide slightly more leverage. Setting $\lambda = 0$ and visually inspecting the bottom panel of Figure SM.17, it is clear that the lower bound of the estimated cumulative effect crosses zero eight years after the initial shock in hybrid regimes, but only after 12 years in democracies. Thus, the positive association between growth and calories lasts about 60% longer in democracies than it does in hybrid regimes. This may be suggestive of the clientelism mechanism B & K introduce, since we would expect the effect of non-programmatic redistribution to fade over time. Yet it is difficult to read too much into this result, given the null findings within and across periods and regime types.

Finally, unlike B & K's model, the general model predicts effect sizes that begin and remain small. Mean daily consumption across the sample is 2,405 calories, so \$100 in GDP growth generates only a 0.5% increase in caloric intake. For a country to move from the

third decile to the cross-national mean in calories consumed, it would require growth of approximately \$1,700 per capita—more than doubling the mean income level. No matter the regime type, it appears that growth engenders negligible redistribution.

Taken as a whole, these findings suggest that there are few cross-regime differences in translating growth into inequality reduction. Hybrid regimes and democracies may be marginally better than autocracies at generating immediate caloric gains for the poor, but effect sizes are small, and grow smaller still over time. In the long run, regime types are essentially indistinguishable in their capacity for pro-poor growth. This interpretation contrasts with that of Blaydes and Kayser (2011), who conclude that “growth in autocracies benefits the poor less than growth in democracies and hybrid regimes” (902). These results suggest that theories of regime transition may need to reconsider the assumption that democracies redistribute more than non-democracies.

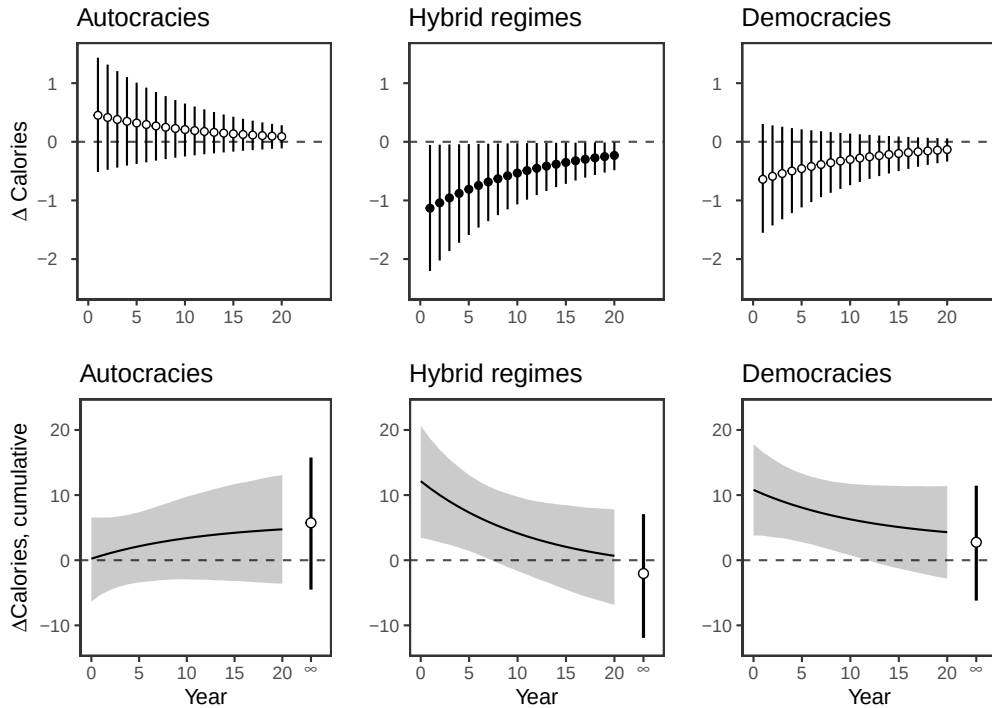


Figure SM. 17: Dynamic relationships between growth and inequality by regime type. The top (bottom) panel plots period-specific (cumulative) effects in average daily per capita calories consumed as the result of \$100 growth in per capita GDP, conditional on regime type, from the full model. LRMs (total effects) are plotted at ∞ . Lines (ribbons) represent 95% confidence intervals constructed from quantiles of 5,000 samples from $\mathcal{N}_{MV}(\hat{\theta}, \mathbb{C}(\hat{\theta}))$.

7.2 Morgan and Kelly (2013)

How can states ensure that the poor benefit from growth? Many studies have uncovered mechanisms through which governments directly redistribute wealth over both the short and long run (e.g., Morley, 2001), including such channels as explicit cash transfers and investments in primary education. Yet very little research is devoted to how states indirectly induce distributive outcomes by shaping market forces.

Morgan and Kelly (2013, “M & K”) investigate these “market conditioning” mechanisms. They note how certain laws and regulations change the economic incentives of private actors—for instance, by inducing monetary stability or encouraging employment of low-wage workers—with knock-on effects for inequality. M & K argue that these market conditioning effects are greatest for human capital spending (HCS), which enhances the skills and productivity of the poor, increasing the value of their labor. Thus, M & K hypothesize that when HCS is low, the rich reap the gains from growth and inequality rises, but when it is high, growth is more efficiently distributed to the poor and inequality declines.

7.2.1 Data and estimation

Data for the response come from the Standardized World Income Inequality Database (Solt, 2009). M & K’s main analysis is based on gross adult-equivalent income adjusted for household composition, which captures pre-tax and -transfer income. From this the authors construct a market Gini index that ranges over 0-100, where larger values indicate a less equal distribution of wealth. For the covariates, M & K use economic growth in real per capita gross domestic product (GDP), measured in thousands of US dollars adjusted for purchasing power parity, and a fifteen-year average of aggregate HCS as a percentage of GDP, both from the Penn World Tables (Heston et al., 2001). Due to data constraints, the sample is restricted to 19 Latin American and Caribbean countries over the period 1980-2000.

Maintaining notation consistent with the general model in Equation SM.1, the authors’

preferred specification is the ECM

$$\Delta y_t = \gamma_0 + \gamma_1 y_{t-1} + \theta_0 \Delta x_t + \theta_2 \Delta z_t + \theta_3 z_{t-1} + \theta_4 \Delta x_t z_{t-1} + \epsilon_t,$$

where y is market Gini, x is GDP, z is HCS, and ϵ is the error term. Note that this model restricts $\theta_1 = \theta_5 = \theta_6 = \theta_7 = 0$, or equivalently, it translates to an ADL where $\beta_1 = -\beta_0$, $\beta_4 = \beta_5 = 0$, and $\beta_7 = -\beta_6$. Using the approach developed above, we solve for the LRM of a shock to GDP, which yields $\frac{-\theta_4 \Delta z_{t+1}}{-\gamma_1}$ (or $\frac{\beta_7 \Delta z_t}{1-\alpha_1}$). Thus, the authors' specification translates into a dynamic system in which both GDP and human capital spending must be *simultaneously* shocked for there to be any conditional relationship over the infinitely long run. This description of the dynamic system contrasts with their theoretical discussion, which predicts that “the *level* of investment in human capital will condition the effect of economic *growth*, or change in GDP” (Morgan and Kelly, 2013, 679, italics in original). Moreover, this specification implies that if such simultaneous shocks do not occur, then the long-run effect of growth on regime change will be zero. These unintuitive implications of the M & K specification follow directly from the parameter restrictions imposed.

We replicate the M & K model before estimating the general model. To ensure comparability, we include the control variables from M & K's favored specification in both.⁷ We also use the authors' data, which centers all variables before estimation, and follow them in clustering standard errors by country.

M & K conduct extensive pretesting of their data, using a mix of unit root and cointegration tests. We extend this pretesting using the same mix of formal tests and visual diagnostics discussed elsewhere in the Supplemental Materials. Again pretesting is complicated by the panel structure, but the evidence indicates that the interaction is a unit root process—in line with both intuition and other analyses of GDP and spending (see, e.g., Box-Steffensmeier et al., 2014). Thus, it is both integrated and—by Proposition 2—cointegrated

⁷These include both lags and differences of legislative partisan balance, inflation, unemployment, and foreign direct investment.

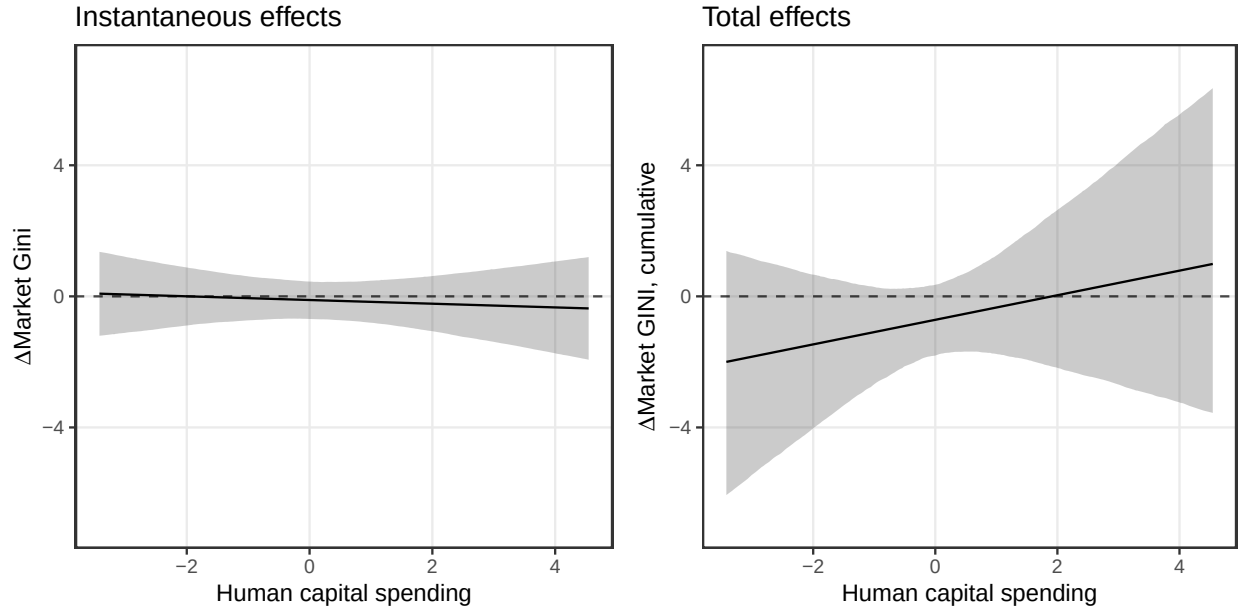


Figure SM. 18: Short- and long-run relationships between growth and inequality, conditional on human capital spending. Lines represent predicted instantaneous effects (left) and total effects (right) on market Gini of mean growth in per capita GDP, holding growth in HCS at its mean value, while varying HCS across its observed range. 95% confidence intervals constructed from quantiles of 5,000 samples from $\mathcal{N}_{MV}(\hat{\theta}, \mathbb{C}(\hat{\theta}))$ are shaded in gray.

with the other variables, which we confirm using the Engle-Granger and Johansen cointegration tests. The evidence indicates that the ECM is an appropriate model for these data.

7.2.2 Results

Results for both models are presented in Table SM.4. Column 2 reproduces the authors' headline findings and column 3 reports estimates for the general model. Most notable among these estimates is that the main finding in M & K disappears in the full model: there appears to be no conditional relationship between growth and HCS on inequality. This result suggests that their model ignores the complex “memory” of the interaction term, inducing bias. Once other cross-time interactions are modeled, only HCS continues to exert a statistically significant effect.

Figure SM.18 visualizes this result. The first panel plots instantaneous effects of economic growth equal to the sample mean (approximately \$63 per capita after the authors'

Table SM. 4: Replication of Morgan and Kelly (2013): comparing the restricted and general models

	M & K model	General model
Market Gini _{<i>t</i>-1}	-0.07* (0.02)	-0.08* (0.02)
ΔGDP_t	-0.04 (0.15)	-0.19 (0.26)
GDP_{t-1}		-0.03 (0.04)
ΔHCS_t	0.74 (1.28)	2.68* (1.90)
HCS_{t-1}	-0.16* (0.06)	-0.19* (0.06)
$\Delta\text{GDP}_t \times \text{HCS}_{t-1}$	-0.24* (0.11)	-0.06 (0.12)
$\text{GDP}_{t-1} \times \Delta\text{HCS}_t$		-0.73 (0.56)
$\Delta\text{GDP}_t \times \Delta\text{HCS}_t$		2.39 (2.14)
$\text{GDP}_{t-1} \times \text{HCS}_{t-1}$		0.03 (0.04)
Observations	197	197
R ²	0.25	0.28
RMSE	1.01	1.01

* $p < .05$. The dependent variable is $\Delta\text{Market Gini}$. Control variables are not reported and standard errors are clustered by country. RMSE is calculated from out-of-sample predictions using 5-fold cross-validation.

centering). Varying the level of HCS across its observed range while holding change in HCS at its mean, we see that growth never has an effect distinguishable from zero. The second panel plots LRMs for the same scenarios; total effects are also indistinguishable from zero. Together, these results suggest that growth does not have an effect on inequality in either the short or long run, irrespective of the level of human capital spending.

Other quantities tell a similar story. Figure SM.19 plots cumulative effects over time, with both the level of and change in HCS fixed at their mean values. At no point in the 20 years following economic growth is there a discernible effect. Moreover, since cumulative

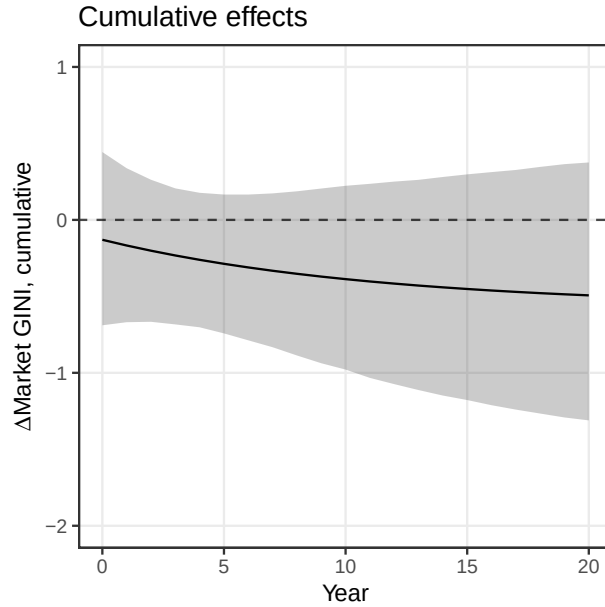


Figure SM. 19: Dynamic relationship between growth and inequality, conditional on human capital spending. The line represents estimates of cumulative change in market Gini over time as the result of mean growth in per capita GDP, holding growth in (and level of) HCS at its mean value. 95% confidence intervals constructed from quantiles of 5,000 samples from $\mathcal{N}_{MV}(\hat{\theta}, \mathbb{C}(\hat{\theta}))$ are shaded in gray.

effects overlap across the entire window, none of the period-specific effects (not plotted) can be distinguished from zero. In short, we cannot be sure of any effect within or across any temporal windows.

Together these findings significantly revise the conclusions in Morgan and Kelly (2013): there is little evidence to support the market-conditioning hypothesis. Yet they also suggest a more subtle relationship between investment in human capital and inequality. While total effects for the mean change in HCS are negative and significant in poor countries, predicted instantaneous effects are positive. Thus, although greater social spending is associated with less inequality in the long run, it actually induces *greater* inequality in the short run. These results suggest that increased HCS comes at a cost which, at least at first, falls on the poorest. Scholars may need to revisit theories of the relationship between redistributive programs and inequality, paying greater attention to the incidence of increased social spending. Whether these patterns are echoed in other contexts remains to be seen,

but either way, inequality in Latin America and the Caribbean appears to be a function of simple redistribution, and not growth as moderated by market conditioning institutions.

7.3 Jennings and John (2009)

What is the relationship between public opinion and government attention to various policy priorities? Jennings and John (2009, “J & J”) study the effect of shifts in public opinion on mentions of policy topics in the Queen’s Speech, a formal statement of the legislative agenda in the United Kingdom (UK). They provide extensive evidence that the variables exist in a dynamic equilibrium. J & J then ask (in their Table 4) if the effect of public attention to macroeconomic issues on Queen’s Speech mentions of economic policy (which we will refer to as simply QS) is moderated by real indicators such as inflation and unemployment.

7.3.1 Data and estimation

Data for the dependent variable come from the UK Policy Agendas Project, which codes the number of mentions of major policy issue areas at the quasi-sentence level for every Speech from the Throne over the period 1911-2012. These data are expressed as an absolute count rather than a percentage of mentions, and so are unbounded (upward). For the purposes of their analysis of the conditional relationship with real economic indicators, J & J focus on the issue area relating to macroeconomics (which they refer to as QS1).

The covariate of interest is the proportion of citizens who believe an economic issue is the most important problem (MIP) facing the UK. The data come from two sources: the King and Wybrow (2001) collection of surveys covering 1937-2000 and the Gallup (2001) original index of surveys over 1959-2001. J & J merge and standardize these data into a single series. To examine economic conditions, J & J study the “misery index,” a simple amalgamation of the inflation and unemployment rates (source not specified).

Maintaining notation consistent with the general model (Equation SM.1), the authors’

specification with an interaction between MIP and misery is the ECM

$$\Delta y_t = \gamma_0 + \gamma_1 y_{t-1} + \theta_6 \Delta x_t \Delta z_t + \theta_7 x_{t-1} z_{t-1} + \epsilon_t,$$

where y is QS, x is MIP, z is misery, and ϵ is error. Note that this model restricts $\theta_0 = \theta_1 = \theta_2 = \theta_3 = \theta_4 = \theta_5 = 0$, or equivalently, it translates to an ADL where $\beta_0 = \beta_1 = \beta_2 = \beta_3 = 0$ and $\beta_4 = -\beta_5 = -\beta_6$. Here the effect of these constraints is to restrict the instantaneous effect of a shock to MIP on QS to $\theta_6 \Delta z_t$, while the LRM reduces to $\frac{\theta_7 z_t + \theta_6 \Delta z_t - \theta_6 \Delta z_{t+1}}{-\gamma_1}$. Together, these quantities suggest that public attentiveness to economic issues affects the legislative agenda *exclusively* through real economic conditions: none of the terms in either quantity is independent of levels of (or change in) the misery index. This implication of the model jars with their theoretical argument, which posits both a direct effect of public opinion on Queen's Speech policy mentions, separate from the interactive effect.

As above, we first replicate the authors' model before estimating the general model. We include J & J's only control variable, an indicator for which party is in government. One observation drops out of both models due to missingness, leaving a time series of just 41 observations.

J & J present a number of tests of residuals for all of the estimated ECMs. We build on this by pretesting each variable for unit roots and visually inspecting ACF and PACF plots. The evidence indicates that each series is integrated. We then examine residuals from the full model, as well as conduct Johansen and Engle-Granger two-step tests to confirm cointegration. While these data are not in a panel, making pretesting slightly more straightforward, the time series is relatively short, diminishing the power of such tests. Again, there is some mixed evidence, but on balance, it points toward the ECM being an appropriate model for these data.

Table SM. 5: Replication of Jennings and John (2009): comparing the restricted and general models

	J & J model	General model
QS_{t-1}	-1.00* (0.17)	-0.93* (0.21)
ΔMIP_t		3.21 (14.76)
MIP_{t-1}		1.51 (7.78)
$\Delta Misery_t$		-0.38 (0.83)
$Misery_{t-1}$		0.09 (0.38)
$\Delta MIP_t \times Misery_{t-1}$		0.12 (0.96)
$MIP_{t-1} \times \Delta Misery_t$		1.04 (1.31)
$\Delta MIP_t \times \Delta Misery_t$	0.44* (0.15)	1.14 (1.66)
$MIP_{t-1} \times Misery_{t-1}$	0.38* (0.10)	0.23 (0.60)
Observations	41	41
R^2	0.49	0.51
RMSE	2.67	3.44

* $p < .05$. The dependent variable is ΔQS . Control variables are not reported. RMSE is calculated from out-of-sample predictions using 5-fold cross-validation.

7.3.2 Results

Table SM.5 presents the results for the J & J model in column 2 and the general model in column 3. Notably, all significant coefficients disappear (except for the control variable for partisanship, which is omitted in the Table). There appears to be neither direct nor conditional relationships between public sentiment, economic conditions, and legislative attention. These null results are illustrated in Figure SM.20. At no level of economic misery does an average shift in public attention yield any effect on legislative attention to economic issues.

Given the Monte Carlo results above, it seems likely that the estimates from J & J's

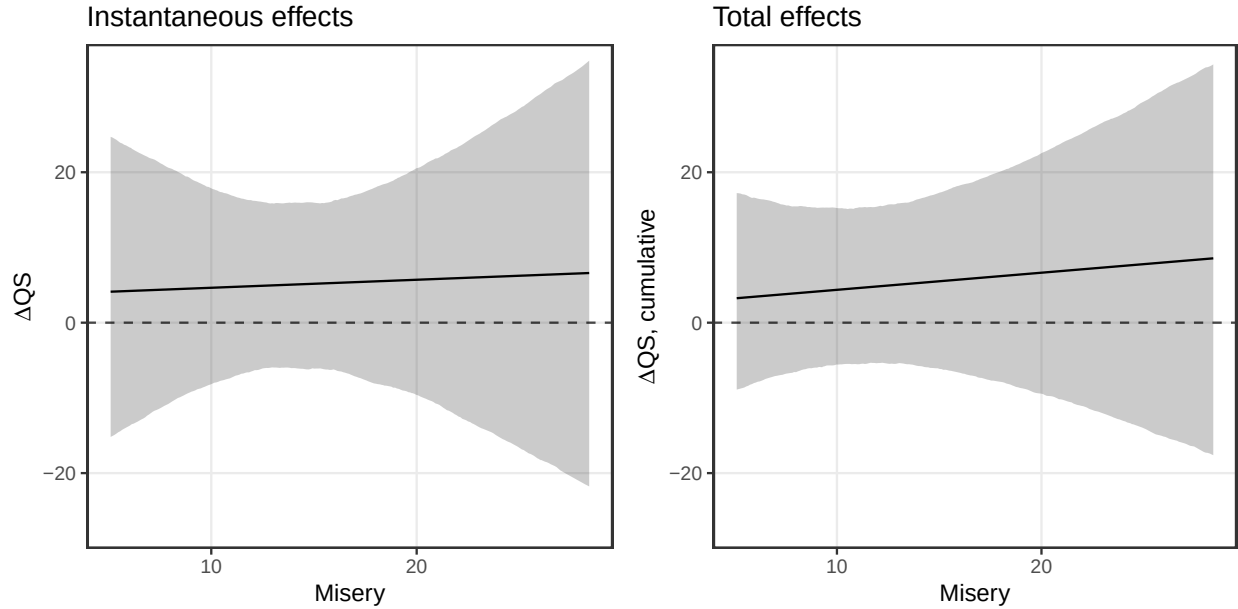


Figure SM. 20: Short- and long-run relationships between public sentiment and policy salience, conditional on the misery index. Lines represent predicted instantaneous effects (left) and total effects (right) on mentions in Queen’s Speeches (QS) as the result of an average increase in public issue salience, holding change in the misery index at its mean value, while varying misery across its observed range. 95% confidence intervals constructed from quantiles of 5,000 samples from $\mathcal{N}_{MV}(\hat{\theta}, \mathbb{C}(\hat{\theta}))$ are shaded in gray.

model suffer from some bias arising from mis-specification. Yet the small sample size, combined with the number of parameters in the general model, suggest that overfitting is a real risk for these data. This interpretation is substantiated by the higher out-of-sample predictive RMSE for the general model. In short, both models likely suffer from bias of some sort. The most reasonable conclusion appears to be that there is simply not enough information in the data to reliably estimate this interaction.

8 Pretesting data

In this section we provide a sample of the pretesting exercises described in the text. In the Blaydes and Kayser (2011) and Morgan and Kelly (2013) applications, the data are organized into panels, and so we present just a small subset of results run separately on each time series, separated by unit (i.e., country). The replication archive contains code to reproduce all visuals and tests of interest. See Box-Steffensmeier et al. (2014) for a thorough guide on interpreting the output of these tests.

8.1 Cavari (2019)

Table SM. 6: *p*-values for univariate unit root tests by variable, Cavari (2019) replication

	ADF test	Ljung-Box test	KPSS test
Performance: economy	0.40	0.00	>0.10
Performance: foreign affairs	0.10	0.00	0.02
MIP: economy	0.10	0.00	0.03
MIP: foreign affairs	0.38	0.00	>0.10
Δ MIP: economy	0.01	0.99	>0.10
Δ MIP: foreign affairs	0.01	0.72	>0.10
Performance: economy $\times$ Δ MIP: economy	0.01	0.74	>0.10
Performance: foreign affairs $\times$ Δ MIP: foreign affairs	0.01	0.92	>0.10
Macro-partisanship (in-party)	0.36	0.00	0.07
Macro-partisanship (out-party)	0.01	0.00	>0.10
Unemployment rate	0.06	0.00	0.09
Δ Unemployment rate	0.01	0.00	>0.10
Divided government	0.03	0.00	0.09
Election year	0.01	0.00	>0.10
Honeymoon	0.01	0.72	>0.10

8.2 Blaydes and Kayser (2011)

Table SM. 7: Univariate ADF tests, Blaydes and Kayser (2011) replication

Variable	Mean p -value	% tests where null rejected
Calories	0.56	0.04
Δ Calories	0.34	0.26
GDP	0.63	0.04
Δ GDP	0.33	0.16
Regime type	0.42	0.00
Δ Regime type	0.51	0.00
GDP $\times$ Regime type	0.71	0.00
Δ (GDP $\times$ Regime type)	0.28	0.57

Table SM. 8: Tests for cointegration, Blaydes and Kayser (2011) replication

Test	Test statistic	p -value
Ljung-Box	100.86	0.00
ADF	-17.39	0.01
KPSS	0.00	0.10
Engle-Granger	-11.07	0.00

Table SM. 9: Johansen test for cointegration, Blaydes and Kayser (2011) replication

λ_{TRACE}				
Null hypothesis	Alt. hypothesis	Test value	95% critical value	90% critical value
$r = 0$	$r > 0$	574.11	53.12	49.65
$r \leq 1$	$r > 1$	372.96	34.91	32.00
$r \leq 2$	$r > 2$	211.60	19.96	17.85
$r \leq 3$	$r > 3$	87.84	9.24	7.52

8.3 Morgan and Kelly (2013)

Table SM. 10: Univariate unit root tests of the interaction term, Morgan and Kelly (2013) replication

Variable	Mean p -value	% tests where null rejected
ADF test	0.55	0.04
Ljung-Box test	0.00	1.00
KPSS test	0.03	0.79
ADF test, first difference	0.34	0.21
Ljung-Box test, first difference	0.26	0.38
KPSS test, first difference	0.07	0.33

Table SM. 11: Tests for cointegration, Morgan and Kelly (2013) replication

Test	Test statistic	p -value
Ljung-Box	27.04	0.13
ADF	-6.23	0.01
KPSS	0.09	0.10
Engle-Granger	-4.98	0.00

Table SM. 12: Johansen test for cointegration, Morgan and Kelly (2013) replication

λ_{TRACE}				
Null hypothesis	Alt. hypothesis	Test value	95% critical value	90% critical value
$r = 0$	$r > 0$	321.07	202.92	196.37
$r \leq 1$	$r > 1$	237.02	165.58	159.48
$r \leq 2$	$r > 2$	173.34	131.70	126.58
$r \leq 3$	$r > 3$	130.05	102.14	97.18
$r \leq 4$	$r > 4$	90.93	76.07	71.86
$r \leq 5$	$r > 5$	61.22	53.12	49.65
$r \leq 6$	$r > 6$	37.84	34.91	32.00
$r \leq 7$	$r > 7$	20.65	19.96	17.85
$r \leq 8$	$r > 8$	6.71	9.24	7.52

8.4 Jennings and John (2009)

Table SM. 13: p -values for univariate unit root tests by variable, Jennings and John (2009) replication

	ADF test	Ljung-Box test	KPSS test
QS	0.42	0.00	0.10
MIP	0.98	0.00	0.05
Misery	0.94	0.00	0.04
MIP $\times$ Misery	0.96	0.00	0.05
Δ QS	0.03	0.07	0.10
Δ MIP	0.01	0.34	0.04
Δ Misery	0.01	0.93	0.10
Δ MIP $\times$ Misery	0.01	0.68	0.10

Table SM. 14: Tests for cointegration, Jennings and John (2009) replication

Test	Test statistic	p -value
Ljung-Box	37.52	0.01
ADF	-3.91	0.02
KPSS	0.31	0.10
Engle-Granger	-4.30	0.00

Table SM. 15: Johansen test for cointegration, Jennings and John (2009) replication

λ_{TRACE}				
Null hypothesis	Alt. hypothesis	Test value	95% critical value	90% critical value
$r = 0$	$r > 0$	47.07	53.12	49.65
$r \leq 1$	$r > 1$	21.83	34.91	32.00
$r \leq 2$	$r > 2$	8.00	19.96	17.85
$r \leq 3$	$r > 3$	0.39	9.24	7.52

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